A Physical Quantification Framework for Crowd Compression Risk: The P-index System and Its Contact-Mechanical Foundation

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Abstract. Quantitative assessment of crowd compression risk has long been limited by the failure of density-based indicators: Tokyo's Yamanote Line and the London Underground routinely reach densities of 6–7 persons/m^2 during peak hours without recorded compressive asphyxia incidents, while the 2022 Itaewon disaster occurred at densities of 8–10 persons/m^2 yet caused 159 fatalities. This paper introduces the P-index, a contact-mechanics-based framework for crowd compression risk quantification that uses a single dimensionless number to simultaneously answer whether an incident will occur at a venue and how severe it will be. The P-index is defined as the total seconds individuals accumulate under above-threshold contact pressure divided by total person-hours, grounded in Hertz contact theory, Coulomb friction, Newton's third law, and the biomechanical hyperbolic lethality envelope. Core innovations include: independently defining axial force direction as the C-value and establishing it as the primary physical variable underlying compressive-asphyxia risk; incorporating psychological factors into the C-value via semi-binary processing to avoid unverifiable continuous psychological coefficients; replacing position-bound density with individual-bound exposure timing τ_i as the basis of risk assessment; establishing a dual-layer reporting structure (P_{baseline} for structural certification, $P_{\max}$ for probabilistic loss estimation); defining venue capacity limit $N_{\max}$ as the phase-transition point of P_{baseline}; and aligning output with industry-standard EP curve format for integration with existing catastrophe model pipelines (RMS, Verisk/AIR, Solvency II). Validation through three public datasets and out-of-sample consistency checking against five historical incidents yields a joint uncertainty of 0.5×–2× the P-index point estimate, providing order-of-magnitude discrimination between operational regimes in the same precision class as physics-based catastrophe models in fat-tail risk domains. The framework adopts a specification-based rather than implementation-based design, consistent with ISO 16730 and Solvency II performance-based regulatory traditions, allowing any simulator capable of outputting individual-level contact-mechanical quantities to serve as the backend.

Keywords: crowd compression risk, P-index, C-value, force chain transmission, individual exposure timing, contact mechanics, EP curve, catastrophe modeling, compressive asphyxia

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1. Introduction

1.1 Density as a necessary but insufficient indicator

Crowd compression risk assessment has historically relied on density (persons/m^2) as the primary indicator. This paper does not dispute that density is a *necessary* condition for compressive risk—compressive asphyxia cannot occur at low density—but argues that density alone is *insufficient* to determine outcome. The determining variable is the cumulative contact-force exposure sustained by an individual, which depends on density together with three additional quantities: the directional coherence of locomotive force across the crowd (the C-value), the geometric configuration of the containing space, and the duration of exposure.

The relationship can be expressed schematically as

$$\text{Effective risk} \;\sim\; \rho \cdot W(C) \cdot G(\text{geometry}) \cdot T(\text{duration}), \tag{1}$$

where ρ is density, $W(C)$ is a non-linear weighting function of the directional-intensity coefficient, G captures the geometric constraints on force-chain formation, and T captures the accumulated exposure time. Density enters as a linear factor; the remaining factors dominate the risk expression in regimes where fatalities occur.

A motorway analogy makes the role of C concrete. At high vehicle density with clear visibility, drivers see brake lights ahead and modulate their inputs in alignment with the actual state of the flow; pile-ups remain rare even when vehicles are closely spaced. At lower density but with poor visibility—fog, a blind crest—a single braking event ahead cascades because downstream drivers cannot align their response to the mechanical state of the flow. The density in the second case is lower; the casualty outcome is worse. What differs is not density: it is the directional coherence of response across the flow. In dense crowds, the corresponding quantity is the C-value, formally defined in Section 4.3.

Density-based regulation, by projecting this multi-factor risk onto a single scalar, discards the information carried by C, G, and T. Any threshold set on ρ alone will therefore produce systematic misclassification: scenarios with high ρ but low $W \cdot G \cdot T$ will be flagged as unsafe when they are not, and scenarios with moderate ρ but high $W \cdot G \cdot T$ will pass the threshold while remaining lethal. The direction of the error is not correctable by adjusting the threshold, because the projection itself is lossy.

Illustrative observations. Reported density values in three well-documented scenarios are broadly consistent with Equation (1), though they are offered here as qualitative illustration rather than quantitative validation. The Itaewon crush of 29 October 2022 (159 fatalities) occurred at a locally reported peak density of 8–10 persons/m^2 in a T-junction geometry that structurally destroys directional coherence. Peak-hour operation of Tokyo's Yamanote Line sustains carriage-average densities of 6–7 persons/m^2 with high directional coherence (uni-directional boarding, shared expectations, short dwell times) and no recorded compressive fatalities across decades of operation. Peak-hour operation of London's Victoria Line sustains 5–6 persons/m^2 under similar coherence conditions. These reported densities are not directly comparable—the

Itaewon figure is a local peak in a stationary trapped population, whereas the metro figures are carriage or platform averages over rapidly circulating individuals—and this incomparability is itself a symptom of density's inadequacy as a stand-alone indicator. Quantitative comparison of these scenarios under the P-index framework appears in Section 1.2, and formal validation is deferred to Section 6.

1.2 Physical comparison of three cities

Applying the P-index framework proposed in this paper (the cumulative seconds per person-hour during which individuals are subject to above-threshold compression) yields the following quantitative comparison:

- **Tokyo Yamanote Line, peak hour:** $P_{\text{baseline}} = 0$. Individual axial contact force is approximately 410 N, well below the lethality threshold (the long-duration lower bound for compressive asphyxia is 1100 N); individual circulation is rapid, with cumulative exposure $\tau_i^{\max} < 10$ s.
- **London Underground, peak hour:** $P_{\text{baseline}} \approx 8.3 \times 10^{-6}$. Individual axial contact force is approximately 620 N, with brief above-threshold excursions in localised regions; the overwhelming majority of individuals experience $\tau_i^{\max} < 30$ s, with no major casualty risk.
- **Itaewon 2022, incident interval:** $P \approx 10^{-3}$–10^{-2}. Individual axial contact force is approximately 5440 N, near the acute lethality threshold; individuals were trapped under bidirectional compression with $\tau_i^{\max} > 100$ s, producing 159 actual fatalities.

The P-index values across the three scenarios differ by at least six orders of magnitude, providing a complete physical account of why comparable densities produce radically different casualty outcomes. Detailed derivation appears in the numerical worked example of Section 4.9.

1.3 The actuarial impasse in insurance and regulation

Liability insurance and regulatory certification for crowd-dense venues currently rely on two foundations: historical claims statistics and density ceiling regulations. The former is constrained by the rarity of compressive incidents, with single-venue annual occurrence rates $< 10^{-3}$, yielding insufficient statistical significance; the latter inherits its engineering basis from fire-evacuation time estimation and has no direct physical connection to compressive asphyxia risk.

Neither approach can answer the core questions of the insurance industry: what is the underwriting threshold for a given venue? How should the base premium be calculated? What is the loss distribution under worst-case scenarios? Mature answers to these questions exist for earthquake, hurricane, and flood catastrophe models—all built on physics-based simulations that output exceedance-probability (EP) curves—but the crowd-risk domain has lacked an equivalent tool.

1.4 Introducing the P-index framework

This paper introduces the P-index framework to fill this gap. The P-index is defined as the cumulative seconds per person-hour during which individuals are subject to above-threshold contact pressure:

$$P_{\text{venue}} = \frac{\sum_{i=1}^{N} \max(0,\ \tau_i^{\max} - \tau_{\text{crit}})}{N \cdot T_{\text{window}}}$$

where $\tau_i^{\max}$ is the maximum cumulative exposure of individual i within the observation window T_{window}, $\tau_{\text{crit}} = 30$ s is the threshold corresponding to the lower bound of "minor injury," and N is the total number of individuals in the venue.

Computation of the P-index rests on four physical quantities: the time–force hyperbolic threshold $F_{\text{thr}}(\tau)$, the directional-intensity composite coefficient C, the force-chain transmission efficiency β, and the force-chain depth L_i. These four quantities already exist in standard contact-mechanical models (such as the Hertz–Coulomb framework); this paper merely promotes them from implicit internal states to explicit, reportable physical outputs.

1.5 Ice and water: two states of the same physical system

The core physical insight of the P-index framework is that the Yamanote Line and Itaewon are not "two different risks" but two states of the same physical system at different parameter values. The underlying contact mechanics, biomechanical thresholds, and force-chain decay laws are identical; the differences reside solely in the values of C (directional intensity) and L_i (force-chain depth).

This relationship admits a phase-transition analogy: the Yamanote Line is ice (stable, low C, shallow L_i), and Itaewon is water (flowing, high C, deep L_i), with C playing the role of temperature. Ice and water have identical chemical composition; they differ only in molecular arrangement. Likewise, the two scenarios share identical physical foundations and differ only in parameter regime.

The strategic implication is that physical parameters relevant to high-risk scenarios can be calibrated through analysis of routine commuter data, because both regimes share the same contact-mechanical basis. This property allows the long-run precision of the P-index framework to improve monotonically with the accumulation of commuter data, unconstrained by the rarity of disasters—providing the framework with a calibration pathway unavailable to traditional statistical models (see Section 6.5).

This shared physical basis is not merely an illustrative analogy but a methodological foundation. Traditional risk assessment attempts to extrapolate from scarce disaster data (approximately 30 recorded major compressive-asphyxia incidents in modern history) to characterise the safety of daily crowd operations (of order 10^9 person-trips per year in a single metropolitan rail system). This paper reverses that direction: the abundant ice-phase data serves as the calibration base that stabilises the physical parameters, and the rare water-phase events serve as verification of the phase-transition boundary, not as calibration targets. The framework therefore does not attempt to predict specific disaster casualty counts from historical incidents; it establishes a baseline distribution from routine data and observes how far historical incidents deviate from that baseline. This methodological inversion is developed in detail in Section 6.5 and underpins the validation strategy of the entire framework.

1.6 Contributions and structure of this paper

The six principal contributions of this paper are:

1. Proposing the P-index as a physically grounded quantitative indicator of crowd compression risk, replacing the failed density indicator;
2. Defining the directional-intensity composite coefficient C with values allowed to exceed 1, encompassing extreme panic scenarios and resolving the structural blind spot of the conventional $[0, 1]$ constraint;
3. Quantifying psychological factors through a semi-binary switching formulation, avoiding the circular definition inherent in continuous panic coefficients;
4. Introducing individual-level cumulative exposure timing τ_i, specifying that it cannot be derived from area statistics, and imposing four proof obligations on any model claiming to output a P-index;
5. Proposing a three-factor decomposition of the effective contact load $F_i = f_{\text{baseline}} \cdot \tilde{C} \cdot (1 - \beta^{L_i+1})/(1 - \beta)$, and clarifying the exposure-computation principle for "passive victims" under path-dependent force chains;
6. Specifying the dual-layer outputs P_{baseline} and P_{max}, the venue physical capacity limit N_{max}, and the sub-region P-index P_Ω, with explicit interfaces to existing catastrophe-model workflows (Solvency II, RMS, Verisk, AIR).

The structure of the paper is as follows. Section 2 defines the formal specification of the P-index framework, including dual-layer outputs, sub-region P-index, EP curves, and N_{max}. Section 3 demonstrates the physical failures of density indicators and existing crowd-dynamics tools. Section 4 establishes the four physical prerequisites for computing the P-index (F_{thr}, C, β, L_i) and the three-factor decomposition $F_i = f_{\text{baseline}} \cdot \tilde{C} \cdot (1 - \beta^{L_i+1})/(1 - \beta)$. Section 5 presents the supporting physics: Hertz contact theory, Coulomb friction, Newton's third law, and graph-theoretic foundations. Section 6 validates the framework through four pathways: physical-consistency tests, public-dataset comparison, reverse-engineering of historical incidents, and cross-scenario extrapolation error analysis—and develops the strategic implication of reverse calibration from subway commuter data. Section 7 demonstrates applications in three domains: insurance actuarial pricing, regulatory certification, and venue design. Section 8 discusses framework limitations and future research directions. Section 9 summarises and concludes.

1.7 Validity conditions of the framework

The P-index framework is defined over populations of *surviving* individuals within the observation window. Its validity conditions reflect a physically important fact: compressive-asphyxia lethality is not governed by a single mechanism across all time scales, but by two mechanistically distinct boundaries separated by an intermediate regime that cannot presently be calibrated by direct experiment. A venue is certified safe under the framework only if it stays clear of *both* boundaries.

Acute-impact boundary (short-duration, structural mechanism). Over short exposures (of order seconds), lethality is driven by structural failure of the thoracic cage—rib fracture and flail chest—at forces of order $F_{\text{acute}} \approx 6000$ N (the upper stage of the three-stage threshold specified in Section 4.2). No individual in the simulation may sustain an instantaneous contact force exceeding this boundary. Individuals crossing it have entered a trajectory toward physiological failure and are, by definition, no longer viable subjects of cumulative-exposure integration; the simulation must terminate integration at the crossing time and reclassify the individual into a separate fatality channel rather than continue accumulating τ_i.

Long-duration boundary (sustained, hypoxic mechanism). Over long exposures (minutes), the operative mechanism is not structural failure but progressive hypoxia from restricted thoracic movement; lethality occurs at substantially lower sustained force (of order 1100 N over several minutes). The individual cumulative exposure $\tau_i^{\max}$ is bounded above by this long-duration lethality threshold. Individuals whose cumulative exposure reaches the bound have transitioned into the fatality regime and no longer contribute to the surviving-population P-index integral, so the venue-level P-index does not grow unboundedly with observation time; it saturates as the surviving population shrinks. This long-duration hypoxic boundary—not the acute structural boundary—is the mechanism operative in the documented crowd-crush fatalities (Hillsborough, Itaewon, Mecca), a distinction developed in the calibration architecture of Section 4.6.

The intermediate regime is deliberately uncalibrated. Between the acute structural boundary and the long-duration hypoxic boundary lies an intermediate time–force regime that cannot be calibrated by controlled experiment on living subjects for ethical reasons. The framework does not claim a precise lethality contour across this interval; it treats the interval as an engineered interpolation between two independently anchored boundaries (Section 4.6) and confines its quantitative claims to order-of-magnitude discrimination between operational regimes. This is a deliberate design choice rather than an unaddressed gap: the two bounding boundaries are each anchored in mechanism-appropriate evidence, while the interior is reported as a relative ordering rather than an absolute contour.

Implication for venue safety certification. Demonstrating that a venue is safe under the P-index framework requires clearing *both* boundaries: no individual exceeds the acute structural boundary at any instant, *and* no individual accumulates exposure beyond the long-duration hypoxic bound over the operating window. Satisfying only one—maintaining low peak force while allowing prolonged sub-threshold compression, or tolerating brief spikes above the acute boundary on the assumption that duration is short—is insufficient. Both conditions follow directly from the two-mechanism structure of the lethality envelope (Section 4.2) and are not additional postulates.

1.8 Scope and limitations

To avoid misinterpretation of the paper's scope, the following items are explicitly outside its remit.

This paper does not replace existing crowd-dynamics tools. FDS+Evac, social force models, cellular automata, and discrete element methods each retain engineering value within their domains of validity. This paper requires only that contact-mechanical quantities be used in the P-index computation step; the other functions of existing tools (path planning, flow prediction, macroscopic statistics) may continue to be used.

This paper does not deny the historical contribution of density indicators. The Fruin Level-of-Service classification, proposed in the 1970s, represented a major advance in crowd safety research at the time. This paper argues only that density is insufficient as a stand-alone indicator in the specific task of distinguishing scenarios such as the Yamanote Line from Itaewon; its continued reference value in low-density evacuation planning is not contested.

This paper does not claim absolute precision. The joint uncertainty of the framework is $\sigma_{\log_{10}} = 0.31$, corresponding to a 95% confidence interval of $0.5\times$ to $2\times$ the point estimate of the P-index. This precision meets industry standards for catastrophe models, but any interpretation that the framework yields "a precise P-index" is a misreading.

This paper does not claim point prediction capability. The framework's output is order-of-magnitude discrimination between distinct physical regimes (routine operation versus incident regime), not precise prediction of casualty counts in a specific future event. Claims such as "the P-index of venue X is 3.2×10^{-3}" should be read as "the P-index of venue X falls in the 10^{-3}–10^{-2} range, distinctly separated from the routine baseline of 10^{-6}–10^{-5}." This precision level is aligned with the industry standard for catastrophe models (± 1 order of magnitude on annual expected loss) and does not require, nor claim to provide, the point-prediction capability demanded of deterministic engineering models.

This paper does not provide complete implementation details. The paper adopts a specification-based rather than implementation-based design: it prescribes what physical quantities a model must output, what consistency tests it must satisfy, and what proofs it must provide, but not how to implement them. Detailed calibration data, implementation optimisations, and commercial case studies are reserved for subsequent commercial reports or consulting deliverables, outside the scope of this paper.

This paper does not address legal liability determination. The framework can provide physical evidence chains for retrospective liability analysis, but specific legal-liability determinations fall within judicial jurisdiction and are outside the discussion of this paper.

2. The P-index Framework: Definition and Formalism

This chapter defines the formal specification of the P-index framework, including the basic definition, comparison with density-based indicators, dual-layer outputs (P_{baseline} and $P_{\max}$), the venue physical capacity limit $N_{\max}$, EP curve output, and the sub-region P-index P_Ω. Detailed development of the physical quantities on which the formal definition depends (τ_i, F_i, F_{thr}, C, β, L_i) appears in Chapter 4.

2.1 Basic definition

Let N denote the total number of individuals in a venue and T_{window} the observation window. For each individual i, define the cumulative asphyxia exposure time $\tau_i(t)$ as the cumulative seconds within $[0, t]$ during which the individual is subject to above-threshold contact pressure. The final maximum exposure of the individual is $\tau_i^{\max} = \max_t \tau_i(t)$.

The P-index is defined as:

$$P_{\text{venue}} = \frac{\sum_{i=1}^{N} \max(0,\ \tau_i^{\max} - \tau_{\text{crit}})}{N \cdot T_{\text{window}}} \tag{2}$$

where $\tau_{\text{crit}} = 30$ s is the critical time corresponding to the lower bound of "minor injury."

The P-index is a dimensionless quantity, interpretable as "the cumulative seconds per person-hour during which individuals are subject to above-threshold compression." $P = 0$ indicates that no individual accumulates above-threshold exposure; $P > 0$ indicates that at least one individual lies within the risk regime.

2.2 C-value ranges and scenario correspondence

Computation of the P-index depends on the directional-intensity composite coefficient C (see Section 4.3). The typical ranges of C and their scenario correspondences are listed below.

C range	Physical state	Representative scenario
0.6–0.8†	Routine egress	Scheduled dispersal, commuter flow
≈ 1.0	Standard emergency evacuation	General fire drill
1.7–1.9‡	Panic with directional convergence	Hillsborough 1989
2.0–2.5	Extreme panic with bidirectional compression	Itaewon 2022, Mecca 2015

Table 1: Typical scenario correspondences for the C-value.

† $C_{\text{baseline}} = 0.6$–$0.8$ corresponds to the "routine egress" regime defined in Section 4.3; the full C-value definition and internal structural decomposition appear in that section.

‡ The point estimate for $C_{\max}$ uses 1.8 (the range 1.7–1.9 reflects the sensitivity-analysis bracket); the broader range 1.5–2.0 is discussed in Section 4.3 for the panic-with-directional-convergence regime. Calibration details appear in the historical-incident back-validation of Section 6.3.

2.3 Supplementary remarks on the formal definition

The construction principle of the P-index numerator is summation of individual-level excess exposure, not regional averaging. The key points are as follows.

Individual-level construction. τ_i is the true cumulative exposure of each individual within the observation window; it cannot be derived from regional statistics such as "total crowded duration in a zone divided by average headcount."

Excess-only contribution. Using $\max(0, \tau_i^{\max} - \tau_{\text{crit}})$ rather than $\tau_i^{\max}$ itself ensures that sub-threshold exposure contributes zero. This design makes $P = 0$ correspond to the unambiguous physical state "no individual has crossed the risk threshold."

Person-hour denominator. The denominator $N \cdot T_{\text{window}}$ is the total person-hours, rendering the P-index comparable across venue scales. A 1000-person venue observed for 1 hour and a 100-person venue observed for 10 hours carry identical denominator weight.

2.4 Dual-layer outputs: P_{baseline} and $P_{\max}$

For any given venue, this framework simultaneously outputs two P-index values.

P_{baseline}**:** The P-index under normal operating conditions (no panic, no external threat, no system failure), corresponding to distribution sampling at $C = C_{\text{normal}}$. Its physical meaning is "structural safety guarantee"—if $P_{\text{baseline}} > 0$, the venue carries asphyxia-event risk even under normal operation.

$P_{\max}$**:** The P-index under worst-case scenarios (panic triggered, directional convergence, bidirectional compression), corresponding to distribution sampling at $C = C_{\text{normal}} + \Delta C_{\text{panic}}$ (effectively C_{panic}). Its physical meaning is "worst-case risk," aligned with the catastrophe scenario analysis used in the insurance industry.

The dual-layer output corresponds to the two-stage pricing logic of the insurance industry: P_{baseline} determines whether coverage is offered (underwriting threshold), while $P_{\max}$ determines the base premium (risk magnitude). See Section 7.1.

2.5 Venue physical capacity limit $N_{\max}$

Define $N_{\max}$ as the maximum venue occupancy at which $P_{\text{baseline}} = 0$. For $N \le N_{\max}$, $P_{\text{baseline}} = 0$ holds identically; once N exceeds $N_{\max}$, P_{baseline} begins to grow nonzero.

$N_{\max}$ is the phase-transition point of P_{baseline} and carries three physical meanings.

Physically grounded capacity. $N_{\max}$ is determined by contact mechanics, directly corresponding to the physical guarantee that "no asphyxia event occurs at this occupancy."

Phase-transition property. P_{baseline} transitions from zero to nonzero at $N_{\max}$, analogous to a thermodynamic phase transition. This property makes $N_{\max}$ a sharp discriminating boundary, not an arbitrary engineering threshold.

Regulatory–insurance alignment. If regulation adopts $N_{\max}$ as the legal capacity and insurance adopts $P_{\text{baseline}} = 0$ as the underwriting threshold, the two are automatically aligned.

See Section 7.2.

2.6 EP curve output

The P-index framework can further output Exceedance Probability (EP) curves, directly interfacing with catastrophe-model frameworks such as Solvency II. The EP curve is generated by Monte Carlo simulation: given a venue with, for instance, $P_{\max} = 2.3 \times 10^{-5}$, 1,000 simulation runs are executed, casualty counts are recorded for each run, and the empirical loss distribution L is obtained by sorting.

The vertical axis of the EP curve is annual exceedance probability; the horizontal axis is loss quantile. Conversion requires the annual event rate λ_{event} (for instance, $\lambda_{\text{annual}} = 80$ for a high-frequency-event venue), from which the empirical loss distribution yields loss quantiles for each return period. This output is fully compatible with earthquake and hurricane catastrophe models and can be integrated directly into existing reinsurance pricing workflows.

2.7 Sub-region P-index P_Ω

For composite venues (multi-purpose stadia, exhibition centres, shopping complexes), a single venue-wide P-index cannot reflect the risk heterogeneity across functional zones. This framework defines the sub-region P-index P_Ω:

$$P_\Omega = \frac{\sum_{i\in\Omega} \max(0,\ \tau_i^{\max} - \tau_{\text{crit}})}{N_\Omega \cdot T_{\text{window}}} \tag{3}$$

where Ω is any spatial sub-region of the venue and N_Ω is the number of exposed individuals within that sub-region.

2.7.1 Calculation principles of P_Ω

P_Ω counts the exposure accumulated by individual i while within sub-region Ω, regardless of whether the individual also accumulates exposure in other sub-regions. A single individual may contribute to multiple P_Ω values simultaneously, but is counted only once in the venue-wide P_{venue}.

2.7.2 Non-additivity of P_Ω

The relationship between the venue-wide P-index and the sub-region P-indices is:

$$P_{\text{venue}} \neq \sum_{\Omega} P_\Omega \tag{4}$$

because the person-hour weights $N_\Omega \cdot T_{\text{window}}$ across sub-regions do not sum to the venue-wide $N \cdot T_{\text{window}}$ (a single individual may traverse multiple sub-regions). This non-additivity is a design feature reflecting the individual-level nature of the P-index.

2.7.3 Applications of the sub-region P-index

The four principal applications of the sub-region P-index (P_Ω) are as follows.

Differentiated pricing. Insurers may zone-price coverage according to each functional area's P_Ω and its proportion of exposed person-hours (Section 7.1).

Retrofit prioritisation. Venue operators may rank retrofit priority by P_Ω, directing budget first to the highest-risk zones (Section 7.3).

Real-time monitoring. Real-time monitoring systems can trigger alerts based on the time-varying $P_\Omega(t)$ of each sub-region (Section 7.3).

Post-incident liability allocation. Following an incident, $P_\Omega(x, y, t)$ provides spatially and temporally precise evidence chains for liability analysis, usable in judicial or insurance-claim proceedings.

2.7.4 Heatmap limit

When the sub-region Ω is taken to the limit of an infinitesimal area element ΔA, P_Ω converges to a spatially continuous function $P_\Omega(x, y)$. Visualisation of this function yields the venue spatial heatmap (Section 6.6), a direct demonstration of the spatial resolution capacity of the P-index framework.

2.8 Chapter summary

This chapter has defined the formal specification of the P-index framework. The P-index is a dimensionless quantity: individual-level cumulative above-threshold exposure summed and divided by total person-hours. The dual-layer outputs P_{baseline} and $P_{\max}$ correspond to underwriting threshold and base premium respectively; $N_{\max}$ is the physical capacity limit at the phase-transition point; EP curves interface with insurance catastrophe-model workflows; the sub-region P_Ω provides spatially resolved risk assessment for composite venues.

Detailed development of the physical quantities on which the formal definition depends (τ_i, F_i, F_{thr}, C, β, L_i) appears in Chapter 4; the physical foundations in Chapter 5; validation in Chapter 6; applications in Chapter 7.

3. Limitations of Existing Methods

This chapter examines the structural physical limitations of existing crowd risk assessment methods. Section 3.1 analyses the failure of density indicators; Section 3.2 examines the historical origin and physical mismatch of per-capita area regulations; Section 3.3 addresses the fundamental problem of position-bound timing; Section 3.4 considers the blind spots of evacuation-time-first design; Section 3.5 analyses the statistical distortion produced by disaster-category aggregation of casualty data.

3.1 Historical context and failure of density indicators

The Level of Service (LOS) classification proposed by Fruin (1971) in the 1970s adopts per-capita area (m^2/person) as its core indicator, dividing crowd density into six grades A through F. This classification became the standard tool for crowd safety assessment over the subsequent half-century, widely applied in building fire codes, event permitting, and structural design.

The LOS classification provides good predictive capacity for crowd flow behaviour in the low-to-medium density range (grades A through D, corresponding to < 2 persons/m^2), reproducing the basic flow–density relationship. However, it fails in the high-density range (grades E and F, corresponding to > 4 persons/m^2): the grade-F density range spans from 4 persons/m^2 (congested but safe commuter conditions) to > 10 persons/m^2 (lethal compressive asphyxia), and within this broad band the density indicator cannot discriminate risk.

The physical reason for this failure is that density measures the instantaneous occupancy state of a venue, whereas the physical basis of compressive asphyxia is the cumulative contact-force exposure sustained by an individual. In the high-density range, contact force and contact-network depth (L_i) are the physical quantities that determine risk; density is only one contributing factor, and its effect is nonlinear.

3.2 Physical mismatch of per-capita area regulations

National fire-code regulations commonly specify legal capacity through per-capita area, with typical values such as 0.46 m^2/person (NFPA 101 standing room, corresponding to approximately 2.2 persons/m^2) or 0.28 m^2/person (dense standing, corresponding to approximately 3.6 persons/m^2).

The historical origin of such regulations lies in fire-evacuation time engineering: ensuring that a venue can be evacuated within a reasonable time under emergency scenarios such as fire. The physical basis is the three-way relationship among flow rate, density, and evacuation time, with no direct physical connection to compressive asphyxia risk.

This mismatch is unobtrusive in low-to-medium density scenarios, since fire-evacuation risk already covers most safety considerations. In high-density scenarios, however, per-capita area regulations may permit a venue configured at 3 persons/m^2 that can transition instantaneously into the 6–10 persons/m^2 incident regime once panic is triggered. Static capacity regulations cannot capture dynamic compression risk.

$N_{\max}$, defined in Section 2.5 as the physical capacity limit derived from contact mechanics, corresponds directly to the physical guarantee of "no asphyxia event occurs" and can serve as a supplement to or replacement for per-capita area regulations.

3.3 The fundamental problem of position-bound timing

Some existing crowd dynamics tools estimate risk via "regional crowded duration," that is, accumulating the time during which density in a venue zone exceeds a threshold. This method has a fundamental physical flaw when distinguishing dynamic flow from static entrapment scenarios.

Consider two scenarios at identical densities.

Scenario A (subway commuting): Venue density holds at 7 persons/m^2 for one hour, but individuals circulate rapidly; each individual passes briefly through the high-density zone (exposure < 10 s).

Scenario B (alley compression): Venue density holds at 7 persons/m^2 for one hour, but individuals are trapped and cannot move; each individual remains continuously in the high-density zone (exposure > 100 s).

The two scenarios yield identical "regional crowded duration" (both equal one hour multiplied by the corresponding area), yet the cumulative contact-force exposure sustained by individuals differs enormously, and the risk outcomes are entirely different (Scenario A is safe, Scenario B is lethal).

This difference is precisely the fundamental physical distinction between the Yamanote Line and Itaewon. Any model adopting position-bound timing is structurally incapable of distinguishing the two scenarios; adding "dynamic correction coefficients" can only fit specific scenarios statistically, without supporting extrapolation.

This framework therefore requires τ_i to be an individual-level physical quantity that cannot be derived from regional statistics (Section 4.4). This requirement is not an implementation choice but a structural response to the basic physical question of whether the Yamanote Line and Itaewon can be distinguished.

3.4 Blind spots of evacuation-time-first design

Traditional evacuation plan design adopts "minimisation of total evacuation time" as its core objective, paired with per-capita area regulation as the second pillar of venue safety design. This design paradigm produced major advances in fire safety from the 1970s through the 2000s, sharply reducing fire-related fatality rates.

However, evacuation-time-first design has two blind spots.

Blind spot 1: Compression risk during evacuation. Evacuation time measures only "when the last person exits the venue" and does not account for the contact-force exposure sustained by individuals during the evacuation itself. Under emergency scenarios such as fire, the evacuation process itself may cause compressive asphyxia incidents (e.g., a portion of casualties in the Station Nightclub Fire 2003 and Kiss Nightclub Fire 2013 arose from evacuation compression rather than from the fire itself).

Blind spot 2: Long-duration risk in non-emergency contexts. Evacuation-time design targets emergency scenarios and provides no corresponding assessment tool for long-duration compression risk under normal operation (e.g., concert dispersal, peak-hour subway commuting). The cumulative exposure time in such scenarios far exceeds typical evacuation times, yet traditional tools cannot quantify it.

The individual cumulative exposure time τ_i in the P-index framework covers both blind spots directly: both the instantaneous contact forces and cumulative exposure sustained during evacuation can be computed, and the cumulative risk of long-duration normal-operation crowding can also be quantified (Section 7.3).

3.5 Disaster-category aggregation of casualty data

Casualty data from historical compressive-asphyxia incidents are frequently aggregated under broader event categories in official statistics, producing systematic distortion.

3.5.1 Typical aggregation patterns

Fire-related incidents. Asphyxia deaths at fire scenes are commonly categorised as "fire fatalities," even when the actual cause is compressive asphyxia during evacuation (e.g., a portion of casualties in the Station Nightclub Fire 2003).

Stampede incidents. The term "stampede" in statistical classification covers multiple causes of death including compressive asphyxia, fall injuries, and cardiopulmonary arrest, rendering specific mechanisms indistinguishable.

Mass events. Casualties in scenarios such as terror attacks, riots, and religious gatherings are commonly assigned to the principal event cause (terror attack, religious conflict), with compressive asphyxia absorbed as a secondary mechanism.

3.5.2 Statistical distortion in large-scale incidents

Official casualty data for large-scale incidents are often under-reported owing to political, religious, or commercial pressures, causing traditional risk models that rely on official statistics to deviate further from the true risk level. The Mecca 2015 incident is a representative case: Saudi authorities reported 769 deaths, while independent compilations reach 2,411 deaths (Associated Press, December 2015) and 2,236 deaths (Agence France-Presse)—a discrepancy exceeding threefold. The Associated Press figure is adopted as the primary reference throughout this paper (see Section 6.3).

Such distortion gives "risk models built on historical casualty statistics" a structural tendency toward underestimation. The P-index framework circumvents this limitation through physics-based modelling: framework calibration does not depend on precise casualty counts but on experimental measurement of contact-mechanical quantities and reverse calibration from large-sample commuter data (Section 6.5).

3.6 Chapter summary

This chapter has identified five structural physical limitations of existing crowd risk assessment methods: the failure of density indicators in the high-density range; the static-evacuation-only scope of per-capita area regulations; the inability of position-bound timing to distinguish the Yamanote Line from Itaewon; the omission of in-evacuation compression risk and long-duration normal-operation risk in evacuation-time-first design; and the statistical distortion induced by disaster-category aggregation of casualty data.

The common root of these five limitations is that existing methods rely on venue-level or zone-level statistics, whereas the physical basis of compressive asphyxia is the individual-level cumulative contact-force exposure. The P-index framework responds to this root issue through individual-level physical computation, developed in detail in the following chapters.

4. Physical Prerequisites for the P-index

4.1 Overview of physical inputs

The P-index computation ultimately reduces to an individual-level time integral: the cumulative seconds $\tau_i^{\max}$ during which individual i is subject to above-threshold contact pressure within the observation window $[0, T]$. For this integral to be meaningfully computed, four physical quantities must be simultaneously defined and output by the simulator.

Quantity	Physical meaning	Section
$F_{\text{thr}}(\tau)$	Time–force hyperbolic threshold	4.2
$C(t)$	Directional-intensity composite coefficient	4.3
β	Force-chain transmission efficiency	4.5
$L_i(t)$	Force-chain depth	4.7

Table 2: The four physical quantities required for P-index computation.

These four quantities combine to produce the effective contact load $F_i(t)$ (Section 4.9); the individual timing logic (Section 4.4), through threshold comparison, then yields $\tau_i(t)$. This chapter develops the definition, measurability, and role within the P-index framework of each of the four quantities in turn.

It should be emphasised that these four quantities are not newly introduced artificial variables but explicit names for, and independent reporting of, physical quantities that already exist within contact-mechanical models. In models such as FDS+Evac (Korhonen et al., 2010), which are based on Hertz–Coulomb contact mechanics, these four quantities are already implicit in the contact force field, the directional components of force, and the topology of contact pairs; this framework merely promotes them from implicit internal states to explicit, reportable physical outputs. Any simulator with contact-mechanical capability can independently compute these four quantities according to the definitions in this chapter and output a P-index without structural modification of its underlying algorithm.

4.2 Time–force hyperbolic threshold

The lethal mechanism of compressive asphyxia is the inability of the thorax to execute respiratory motion under sustained external force, leading to hypoxic unconsciousness and cardiopulmonary failure. The biomechanics literature establishes clearly that this mechanism is determined jointly by force magnitude and duration of application—it is neither a pure force threshold nor a pure time threshold, but a hyperbolic relation between the two.

4.2.1 Summary of literature data

The following table summarises representative data from the biomechanics and forensic pathology literature concerning thoracic compressive lethality (Kroll et al., 2017).

Thoracic force (N)	Time to lethality	Description
∼6,227	∼15 s	Acute impact
∼4,050 ± 320	30–60 s	Dynamic compression
∼2,550 ± 250	60–180 s	Static compression
∼1,112	4–6 min	Long-duration low pressure
∼3,000	instantaneous	Rib fracture (structural failure)

Table 3: Representative force–time data for thoracic compressive lethality, tabulated from the biomechanics and forensic pathology literature as synthesised by Kroll et al. (2017). The mechanism-specific anchoring of each row—acute-impact anchor for the top row (6227 N, structural failure), long-duration lethal anchor for the fourth row (1112 N, sustained hypoxia)—is developed in Section 4.6. Rows 2–3 (4050 N dynamic, 2550 N static short-duration) lie within the engineered interpolation region between the two anchors; they are used as intermediate reference points for the hyperbolic envelope fit rather than as independent anchor points. Row 5 (3000 N rib fracture) is a structural-injury reference threshold, not part of the lethality envelope. Independent evidence streams supporting each anchor are documented in Section 4.6.

The rows of Table 3 span four physically distinct thoracic loading mechanisms rather than a single continuous curve. The top row (∼6,227 N, ∼15 s) is governed by acute structural failure of the thoracic cage and is anchored in the biomechanical and cadaveric-impact literature. The fourth row (∼1,112 N, 4–6 min) is governed by prolonged hypoxia from restricted thoracic movement and is anchored independently in sand-burial fatality series and forensic autopsy series of crush asphyxia. Section 4.6 sets out the mechanism-specific evidence streams for each anchor; the present section uses the tabulated values as inputs to the hyperbolic-envelope interpolation developed in the next subsection without prejudging the anchoring analysis.

It should be noted that "static" and "dynamic" in the table refer to the mode of force application—static for sustained, steady pressure (e.g., body stacking) and dynamic for intermittent impact pressure (e.g., crowd surges). In actual crowd compression events, the force field sustained by individuals is typically a mixture of both, with the static mode dominant.

4.2.2 Hyperbolic envelope approximation

The four reference points tabulated above provide anchor values for thoracic compression lethality at distinct time scales, drawn from biomechanical measurements and historical case synthesis (Kroll et al., 2017). We propose a phenomenological hyperbolic envelope connecting these anchor points:

$$F \cdot \tau^{0.6} \approx K_{\text{lethal}}, \quad K_{\text{lethal}} \approx 1.5 \times 10^4 \text{ N} \cdot \text{s}^{0.6} \tag{5}$$

The exponent 0.6 is obtained by log-linear interpolation across the four anchor points and yields $R^2 > 0.95$ when the anchors are treated as a fitting target. We emphasise that this hyperbolic form itself is an *original phenomenological hypothesis proposed in this work*, not a result established by Kroll et al. (2017) or any other single cited source: the referenced literature provides isolated static and dynamic force thresholds but does not establish a continuous time–force trade-off curve. Analogous dose–response envelopes are well established in adjacent biomechanical domains (notably the Head Injury Criterion and thoracic viscous criterion in automotive safety), and the present hypothesis extends this general methodological tradition to the crowd-compression context. Independent experimental validation of the specific exponent and coefficient values remains open work.

The physical interpretation is that a near-power-law trade-off exists between the force required for lethality and the time required: higher force shortens the time to lethality, lower force extends it, but with a lower bound (below approximately 1,000 N, lethality no longer applies regardless of duration).

The exponent 0.6, slightly less than 1, indicates that the lethal effect of cumulative dosage is sub-linear in time—distributing the same cumulative dose (force × time) over a longer period produces slightly less lethal effect than concentrating it over a short period. This is consistent with thoracic physiological recovery mechanisms (short-duration high pressure produces greater irreversible damage; long-duration low pressure allows more compensatory capacity).

4.2.3 Three-stage engineered threshold

While the hyperbolic envelope provides the more precise expression, a continuous threshold function complicates integral computation and reduces auditability in engineering practice. The framework therefore adopts a three-stage step-function threshold as the operational definition:

$$F_{\text{thr}}(\tau) = \begin{cases} 6000\ \text{N}, & \tau < 15\ \text{s} \\ 2500\ \text{N}, & 15 \leq \tau < 60\ \text{s} \\ 1100\ \text{N}, & \tau \geq 60\ \text{s} \end{cases} \tag{6}$$

The three stages correspond to the three time intervals where literature data points cluster:

- Under 15 s: acute compressive lethality (heavy-object collapse, large instantaneous impact).
- 15–60 s: dynamic compressive lethality (crowd surges, accumulated push impacts).
- Over 60 s: static compressive lethality (body stacking, prolonged retention in a force field).

This step-function treatment preserves the physical essence while making the integral computation discrete, the results reproducible, and the threshold applicable to each individual at each instant traceable by external auditors.

4.2.4 Three-level injury classification

Corresponding to the above threshold, the final injury outcome of each individual is classified into three levels by $\tau_i^{\max}$ and peak F_i:

Light: $\tau_i^{\max} \geq 30$ s and peak $F_i < 2500$ N. Possible loss of consciousness with recovery potential and no permanent organ damage.

Severe: $\tau_i^{\max} \geq 60$ s and $F_i \geq 2500$ N, or $\tau_i^{\max} \geq 180$ s and $F_i \geq 1100$ N. High probability of permanent organ damage requiring immediate medical intervention.

Fatal: Individual sustains F_{thr} for the corresponding interval—specifically, $F_i \geq 6000$ N persisting to $\tau = 15$ s, or $F_i \geq 2500$ N persisting to $\tau = 60$ s, or $F_i \geq 1100$ N persisting to $\tau \geq 240$ s. These three conditions are obtained by substituting three discrete time points into the hyperbolic envelope of Section 4.2, and are continuous with each other on the envelope.

These three levels carry different casualty weights in EP curve generation (see Section 2.6 and Chapter 6), allowing the final loss estimate to reflect the actual injury structure rather than a simple "deceased/survived" binary.

4.3 C-value: directional-intensity composite coefficient

The C-value is the single physical quantity in this framework most strongly correlated with the P-index, and it serves as the coupling intermediary for the other physical quantities in this chapter (β, L_i, τ_i). This section develops the formal definition of the C-value, its physical interpretation, and the semi-binary switching quantification of psychological factors.

4.3.1 Formal definition

Let $f_{\text{raw},i}(t)$ denote the instantaneous push force of individual i at time t—corresponding to the self-driving force term $m_i(v_i^0 - v_i)/\tau_{\text{relax}}$ in models such as FDS+Evac (Korhonen et al., 2010). Let $f_{\text{axial},i}(t)$ denote the component of this push along the axis of the local primary force chain. The C-value is defined as:

$$C_i(t) = \frac{f_{\text{axial},i}(t)}{f_{\text{baseline}}} \tag{7}$$

where $f_{\text{baseline}} = 200$ N is a physiological reference value corresponding to "the push force exerted by a normal adult during purposeful, brisk motion without panic." This reference value is derived from human-factors literature on door-push forces and transfer-walking forces—a typical door-push force of 250 N multiplied by a normative directional alignment ratio of approximately 0.8 yields 200 N as the normative baseline for the actual axial component.

4.3.2 Why the C-value can exceed 1

The denominator of the C-value is a fixed reference value rather than a maximum, so the C-value is not restricted to the $[0, 1]$ range. When an individual's push intensity rises (panic, urgency, external threat) or directional alignment increases (unidirectional flow, axial convergence of bidirectional compression), $f_{\text{axial},i}$ can substantially exceed 200 N, making the C-value greater than 1.

The complete operational range of the C-value is as follows.

C range	Physical state	Representative scenario
0.2–0.4	Sparse contact	Subway car under normal flow
0.5–0.8	Routine egress	Scheduled dispersal
≈ 1.0	Standard emergency evacuation	General fire drill
1.5–2.0	Panic with directional convergence	Hillsborough 1989, Love Parade 2010
2.0–3.0	Extreme panic with bidirectional compression	Itaewon 2022, Mecca 2015

Table 4: Complete operational range of the C-value.

Any model that restricts the C-value (or an equivalent quantity) to the $[0, 1]$ range is structurally incapable of reproducing the historical incidents with $C \geq 1.5$ tabulated above. This restriction is not a matter of fitting precision but of structural blindness—such models, even when supplemented with an additional "panic coefficient," can only fit normal scenarios within the $C \leq 1$ range and cannot extrapolate to the lethal regime. The framework's open design for the C-value is a necessary condition for its admission into insurance actuarial workflows.

4.3.3 Internal structure: direction × intensity decomposition

While the C-value is reported externally as a single composite number, it can be decomposed internally as the product of two independent physical quantities:

$$C_i(t) = C_{\text{dir},i}(t) \cdot C_{\text{int},i}(t) \tag{8}$$

where:

- $C_{\text{dir},i}(t) \in [0, 1]$ is the directional alignment coefficient: the ratio of an individual's push force aligned with the local primary force-chain axis. $C_{\text{dir}} = 1$ when fully aligned, $C_{\text{dir}} = 0$ when fully perpendicular.
- $C_{\text{int},i}(t) = f_{\text{raw},i}(t)/f_{\text{baseline}} \in [0, \sim 3]$ is the intensity coefficient: the ratio of raw push force to the baseline. Under normal conditions $C_{\text{int}} \approx 1$; under extreme panic it can reach 2.5–3.0.

The physical interpretation is that an increase in the C-value can arise from directional convergence (rising C_{dir}), from increased push intensity (rising C_{int}), or from both simultaneously. The typical pathway in historical incidents is both at once—panic triggers an increase in push intensity, while geometric dead-ends force directional convergence, allowing the C-value to escalate from the normative 0.5–0.8 to the lethal 1.8–2.2.

The single-number composite C-value is retained as the external reporting interface, but the internal structure must be complete: any model that can output only a single C-value without independently providing the decomposition into C_{dir} and C_{int} cannot pass the physical consistency tests (Chapter 6).

4.3.4 Semi-binary switching quantification of psychological factors

Psychological variables such as crowd panic, agitation, and urgency are treated in traditional crowd safety literature as "intensity grades" or "behavioural parameters," for example by quantifying panic as a continuous parameter in $[0, 1]$ multiplied into kinematic quantities such as push force or velocity. This treatment has two fundamental problems at the physical level.

Problem 1: Psychological variables are not externally observable. Researchers cannot directly measure "an individual's level of panic" at an incident site, only retrospectively infer it from behaviour (velocity, push force, direction). This renders psychological parameters circularly defined—panic explains behaviour, and behaviour defines panic.

Problem 2: Continuous parameters lack calibration anchors. Differences such as "panic coefficient 0.7 vs. 0.8" cannot be externally validated or calibrated, ultimately reducing to free parameters of model fitting.

The framework adopts semi-binary switching to circumvent both problems. The specific approach is as follows.

Step 1. Establish two clearly measurable reference states:

- **Normal state:** scenarios with dispersed direction and low push intensity. This can be calibrated from extensive observational data on subway commuting, normal-operation shopping centre flows, etc., yielding the typical C-value distribution under this state (denoted C_{normal}).
- **Panic state:** scenarios in which the crowd behaviour falls within the interpolated C_{panic} region defined by the calibration architecture of Section 4.6. This region is bounded below by the empirically calibrated C_{normal} distribution and above by the biomechanically anchored F_{acute} threshold, with the intermediate range engineered by interpolation rather than directly calibrated from historical incident data. Historical incidents such as Hillsborough, Love Parade, and Itaewon serve as out-of-sample consistency checks against this interpolated range (Section 6.3), not as calibration sources for the C_{panic} distribution itself. The typical C-value range associated with this state, denoted C_{panic}, is therefore an interpolated engineering estimate carrying the directional conservativeness discussed in Section 4.6.

Step 2. Define the physical impact of panic as the difference between the two states' C-values:

$$\Delta C_{\text{panic}} = \langle C_{\text{panic}} \rangle - \langle C_{\text{normal}} \rangle \tag{9}$$

This difference is the difference of two measurable distributions and is itself a measurable quantity—it does not depend on any "panic coefficient" fit and does not require assumptions about the intrinsic intensity grade of panic.

Step 3. Apply the binary switch in risk assessment: under normal operation, use C_{normal} to compute P_{baseline}; under worst-case scenarios, use $C_{\text{normal}} + \Delta C_{\text{panic}}$ (effectively C_{panic}) to compute P_{max}.

This treatment is termed “semi-binary” because the final C-value remains continuous (it can take any value between C_{normal} and C_{panic}, corresponding to partial-panic scenarios), but the psychological variable itself enters only in binary form—either belonging to the normal statistical distribution or to the panic statistical distribution, with no intermediate “panic intensity 0.7” ambiguous state.

4.3.5 Advantages of semi-binary treatment

This treatment offers three explicit advantages over traditional continuous-parameter approaches.

First, all input quantities are externally verifiable. C_{normal} is calibrated from massive subway commuter data (over 10^7 person-hour observations daily), and C_{panic} is calibrated from historical incident databases (see Section 6.3). Both are measurement results that external auditors can independently reproduce, not internal fitting parameters.

Second, the circular definition of psychological parameters is avoided. The framework does not need to answer the unmeasurable question “what is the individual’s level of panic,” but only the measurable question “does the crowd behaviour in this scenario belong to the panic statistical distribution.” The former is a psychological controversy; the latter is a statistical discrimination.

Third, natural correspondence with the $P_{\text{baseline}}/P_{\text{max}}$ dual-state structure. The two anchor points C_{normal} and C_{panic} become directly the physical foundation of the dual P-index system—P_{baseline} uses distribution sampling from C_{normal}, and P_{max} uses sampling from C_{panic}. The “structural safety vs. worst-case” dual interpretation is thus carried directly by physical quantities rather than being merely an engineering distinction.

4.3.6 Computational freedom for the C-value

The definition of the C-value is the physical relationship “axial push force divided by baseline,” but the specific algorithm for computing $f_{\text{axial},i}$ is not prescribed by this framework. Any simulator with contact-mechanical capability may:

- Project the net force on an individual onto the local force-chain axis directly;
- Derive it from the velocity and acceleration fields;
- Estimate the axial component from the topology of the contact network.

So long as the C-value output by the simulator reproduces the C_{panic} range of historical incidents in the physical consistency tests (Section 6.1) and reproduces the C_{normal} range in subway commuter observational calibration, the algorithm is deemed compliant.

This open design has two strategic purposes. First, it avoids binding the framework to a single algorithmic implementation, facilitating adoption of the C-value system by existing simulation platforms without rewriting their core. Second, it leaves the optimisation space at the commercial implementation level (contact-detection acceleration, numerical stability of axial projection, etc.) as a differentiating competitive domain among simulators, not part of the framework's open specification.

4.4 Individual timing logic and proof obligations

The P-index computation rests on the independent timing quantity τ_i of each individual. This section specifies the formal definition of this timing quantity, the conditions for its determination, and the proof obligations that any model claiming to output a P-index must satisfy.

4.4.1 Formal definition

For each individual i in a venue, the cumulative asphyxia exposure time $\tau_i(t)$ is defined as the cumulative seconds within $[0, t]$ during which individual i is subject to above-threshold contact pressure.

"Above-threshold contact pressure" is determined by the hyperbolic envelope of Section 4.2—the combination of the current effective contact force F_i sustained by the individual and the current τ_i falls within the lethal region.

The final injury outcome of the individual is determined by $\tau_i^{\max} = \max_t \tau_i(t)$:

- $\tau_i^{\max} \geq 60$ s: An asphyxia event exists in the venue, with individual i as a party to the event.
- $\tau_i^{\max} \geq 60$ s with high peak F_i: Individual i enters the severe or fatal injury regime (Section 4.2 classification).

The numerator of the P-index is the sum of excess values $\max(0, \tau_i^{\max} - \tau_{\text{crit}})$ for all individuals with $\tau_i^{\max} \geq \tau_{\text{crit}} = 30$ s (see Section 2.3). Therefore "venue P-index greater than zero" and "at least one individual's cumulative exposure exceeds the threshold" are mathematically equivalent.

4.4.2 Individual independence requirement

τ_i is an individual-level physical quantity that cannot be derived from regional statistics. Specifically:

- Cannot use "total regional crowded duration divided by average headcount" as an estimate of individual τ_i
- Cannot use "bottleneck dwell time" as the τ_i for all individuals passing through that bottleneck
- Cannot use any form of spatial or temporal averaging in place of the true accumulation along an individual's trajectory

The model must be able to report the current τ_i value for every individual in the venue at every time step. Any computation result based on regional statistical approximations, even if numerically close to the true P-index, is deemed non-compliant under the physical consistency tests (Chapter 6).

The physical justification for this requirement has been argued in detail in Section 3.3—position-bound timing cannot distinguish dynamic flow (each individual passes briefly) from static entrapment (the same individual is continuously pressed), and this is precisely the fundamental difference in risk nature between the Yamanote Line and Itaewon. Reverting to position-bound timing within the P-index framework would be a physical regression to the failed pathway of the density framework.

4.4.3 Treatment of discontinuity and recovery

Exposure of individuals to high-pressure environments is typically not a single continuous event but consists of multiple contact/release cycles. For example, an individual may sustain 20 s of high pressure during one surge, then release for 10 s, then enter another 25 s of high pressure—in such cases, what should the final value of τ_i be?

The framework specifies as follows.

First, short releases cannot directly reset to zero. The physiological state of an individual who has experienced significant compression does not instantaneously recover upon release—thoracic tissue stress, cardiopulmonary compensation depletion, and post-hypoxia metabolic debt all require substantial time to recover. Therefore τ_i for an individual who has just undergone 20 s of compression and then released should not immediately return to zero; otherwise the model will systematically underestimate the true risk from accumulated multiple compressions.

Second, fully ignoring brief compression is also unacceptable. The model must not set a threshold such as "compression events shorter than X seconds are not counted"—this produces discontinuous jumps at the boundary and artificially eliminates long-duration low-intensity exposure.

Third, reasonable recovery mechanisms are permitted. Without directly resetting to zero or ignoring brief events, the model may adopt any recovery treatment consistent with physiological reality—e.g., recovery rate dependent on current τ_i state, recovery time constant dependent on pressure exposure history, or other mechanisms supported by biomechanics literature. The framework imposes no specific recovery algorithm but requires that the output reproduce expected cumulative behaviour in validation testing.

Fourth, the net effect of accumulation and recovery must make the time for a single continuous high-pressure exposure to reach the lethal threshold consistent with the biomechanics literature. That is, under maximum pressure, the true time for an individual to accumulate from $\tau_i = 0$ to $\tau_i = 60$ s should fall within the lethal-time range given by the biomechanics literature (Section 4.2).

4.4.4 Proof obligations

Any model claiming to output a legitimate P-index must provide the following proofs in its report.

Proof 1: Complete presentation of the individual τ_i distribution. The report must include the complete statistical distribution of $\tau_i^{\max}$ for all individuals in the venue (e.g., histogram or empirical CDF), not merely a single P-index number. The shape of this distribution directly reflects the correctness of the timing logic—dynamic flow scenarios should exhibit short-tailed distributions, and static risk scenarios should exhibit long-tailed distributions. A model reporting only a single P-index number while refusing to disclose the individual τ_i distribution cannot be verified by external auditors against the individual independence requirement of Section 4.4.

Proof 2: Non-averaging proof. For any individual reported with $\tau_i^{\max} \geq 60$ s, the model must reconstruct the complete $\tau_i(t)$ trajectory of that individual—the cumulative τ_i value of that individual at every instant within the time window. This trajectory proves that the 60 s is not an artefact of "distributing the total venue crowded seconds evenly across all heads" but the true accumulation of that specific individual.

Proof 3: Discontinuity treatment proof. If an individual's $\tau_i^{\max}$ arises from accumulation across multiple non-contiguous exposure events, the model must separately present each event's start time, duration, peak force, and recovery interval. This decomposition proves that the model's recovery treatment complies with Section 4.4—neither directly resetting to zero nor accumulating without limit.

Proof 4: Threshold lethal-time compatibility. The model must reproduce the lethal-time range from the biomechanics literature in a standard test scenario (single continuous exposure of a single individual under maximum pressure). The specific execution protocol is specified in the physical consistency tests of Section 6.1.

The design purpose of these four proof obligations is to ensure that the P-index computation process can be traced and verified step-by-step by external auditors. A model that cannot provide any one of the above proofs should not have its output P-index accepted by regulatory or actuarial workflows, however numerically reasonable the value may appear.

4.4.5 Structural requirements on existing models

The specifications of Section 4.4 imply a series of structural requirements on model architecture.

First, individual-level state variables. The model must maintain an independent timing state for each agent, with that state continuously updated through time-step evolution. Models that maintain only kinematic quantities such as position and velocity, and infer τ_i through post-processing regional statistics, cannot meet the individual independence requirement.

Second, individual-level force-field input. The model must compute the effective contact force $F_i(t)$ sustained by each agent at each time step. This requires the model to possess the computational capacity for the contact network (Section 4.5) and force-chain depth (Section 4.7)—traditional models with only two-body repulsion cannot directly output F_i.

Third, state-dependent threshold determination. F_{thr} varies with the current τ_i (the three-stage hyperbolic threshold of Section 4.2), and the model's numerical integration logic must handle state-dependent threshold switching. A simplified treatment with a fixed threshold introduces systematic bias at the threshold switching point.

Fourth, continuous evolution capability. The accumulation and recovery of τ_i are continuous processes; the model must perform integration at a sufficiently small time step (typically $\Delta t \leq 0.1$ s). Coarse time-step integration introduces artificial bias between accumulation and recovery, distorting the individual τ_i trajectory.

These four requirements are not local adjustments that can be achieved by feature extensions but basic specifications for the model's core data structures and time integration logic. Any model lacking any of these four capabilities cannot produce a legitimate P-index consistent with this framework—even if it outputs a number and labels it "P-index," that number cannot reproduce the framework's core predictions in physical consistency testing, nor can it pass the proof obligation checks of Section 4.4.4.

4.5 Force-chain transmission efficiency β

In dense crowds, contact forces do not act in isolation between adjacent individuals; they propagate through chains of bodies in mutual contact, analogous to force chains in granular media (Cates et al., 1998; Majmudar and Behringer, 2005). The transmission efficiency $\beta \in (0, 1)$ quantifies the fraction of force retained when transmitted from one body to the next along such a chain.

4.5.1 Physical definition

Let f_k denote the contact force exerted on the k-th body in a force chain initiated by an upstream pushing force f_0. The effective load accumulated at body i, located at depth L_i in the chain, is given by:

$$F_i^{\text{eff}} = \sum_{k=0}^{L_i} \beta^k f_k \tag{10}$$

When the upstream forcing is approximately uniform ($f_k \approx f_{\text{baseline}} \cdot \tilde{C}$), Eq. (10) reduces to the closed-form geometric series:

$$F_i^{\text{eff}} = f_{\text{baseline}}\, \tilde{C}\, \frac{1 - \beta^{L_i+1}}{1 - \beta} \tag{11}$$

4.5.2 Numerical value and provisional calibration

At present, the value of β adopted in this framework rests on analogical reasoning from adjacent physical systems and inference from the closest available human-column experiment, rather than on direct calibrated measurement of force transmission across stacked human subjects. Two lines of evidence support the working range $\beta \in [0.95, 0.99]$:

- **Granular-physics analogy.** Per-layer force retention in dense granular systems has been measured in the range 0.85–0.97 (Liu et al., 1995; Majmudar and Behringer, 2005). Because human bodies are more elastic and compliant than rigid granular particles, and because the axial contact area between torsos is substantially larger than that between rigid grains, a slightly higher retention factor for human contact chains is physically expected.

- **Human push-propagation experiments.** Feldmann and Adrian (2023) report controlled push transmission along columns of standing human subjects, providing the closest available direct evidence. Their attenuation profiles are consistent with per-layer retention in the upper part of the granular range when reinterpreted through the per-layer decomposition of Eq. (11). Direct calibration against a broader set of human-column configurations remains open work.

Throughout this paper we adopt the provisional value

$$\beta = 0.978 \tag{12}$$

positioned at the upper end of the plausible range. This choice maximises predicted force accumulation (and hence risk) for a given chain length L_i, biasing the P-index toward risk over-estimation rather than under-estimation, in line with the conservative-calibration principle of Section 4.6. Precise experimental calibration of β for human contact chains is identified as future work (Section 6.5 and Chapter 8); the sensitivity analysis of Section 6.4 confirms that framework outputs remain within the order-of-magnitude discrimination band across $\beta \in [0.95, 0.99]$.

4.5.3 The 80%/10-layer rule

A practical implication of $\beta = 0.978$ is that the cumulative load saturates rapidly. Substituting into Eq. (11):

$$\frac{F_i^{\mathrm{eff}}(L_i = 10)}{F_i^{\mathrm{eff}}(L_i \to \infty)} = 1 - \beta^{11} \approx 0.78 \tag{13}$$

That is, roughly 80% of the maximum possible chain load is attained within the first 10 bodies. Beyond this depth, additional chain members contribute marginally. This justifies a practical truncation rule:

$$L_i^{\mathrm{eff}} = \min(L_i, L_{\mathrm{max}}), \quad L_{\mathrm{max}} = 12 \tag{14}$$

When L_i exceeds 12, the effective depth is capped at 12. This bound is conservative (it slightly underestimates accumulated load for very deep chains but the error is bounded by $\beta^{13} \approx 0.75\%$ of F_∞) and prevents pathological growth in numerical implementations where contact-graph noise could otherwise produce spuriously long chains.

4.6 Calibration architecture and the burden-of-proof threshold

The four physical quantities defined in Sections 4.2–4.7 are calibrated through an anchor-and-interpolation architecture rather than through direct fitting to crowd incident casualty data. This section makes the architecture explicit, identifies the three mechanism-distinct anchor points on which it rests, addresses the independence of those anchors, and derives an operational consequence: the position of two burden-of-proof boundaries—one for acute impact, one for long-duration compression—in venue safety certification.

4.6.1 Three mechanism-distinct anchors and the interpolation regions

The C-value operational range specified in Section 4.3 (Table 4) and the three-stage threshold function of Eq. (6) together rest on three independent calibration anchors corresponding to three physically distinct thoracic loading mechanisms, with explicitly interpolated regions between them. The framework does not fit a single continuous lethality curve across mechanisms; each mechanism is anchored by mechanism-specific evidence, and only the regions between anchors are engineered through interpolation.

Routine-load anchor: C_{normal}. The distribution of C-values under routine commuter conditions is calibrated from large-scale observational data of urban rail systems (of order 10^9 person-trips per year at metropolitan scale). This anchor is independent of any historical crowd incident data. Detailed calibration methodology appears in Section 6.5.

Acute-impact anchor: F_{acute}. The acute-lethality force threshold ($\approx$6000 N at $\tau < 15$ s, corresponding to the flail-chest structural-failure mode of the thorax) is drawn from experimental and biomechanical modeling sources deliberately unrelated to crowd incident reconstruction: biomechanical modeling of thoracic structural failure (Kroll et al., 2017), cadaveric impact testing from automotive safety research (Kroell et al., 1974; Kent et al., 2004), historical judicial pressing records (McKenzie, 2005), and vending-machine tipping fatality reports (Cosio and Taylor, 1992). None of these sources involve reconstruction of crowd compression events. The validation domain of this anchor is confined to short-duration, structural-failure loading; Kroll et al. (2017) explicitly state that their biomechanical model "says nothing about long-term > 4 minute moderate chest compression crowd crush," where the fatality mechanism is prolonged hypoxia rather than flail chest. This anchor therefore does not govern the Hillsborough, Love Parade, Mecca, Astroworld, or Itaewon incidents, all of which are long-duration hypoxic events falling outside its stated scope.

Long-duration lethal anchor. The lethal boundary for prolonged sub-flail-chest thoracic loading ($F \sim 1{,}100$ N sustained for ≥ 4 min, corresponding to the long-duration row of Table 3) is anchored by two independent evidence streams whose failure mechanism is purely respiratory restriction, not structural rib fracture. *Sand-hole and sand-burial fatality series* (Maron et al., 2007) document cases in which the applied pressure can be estimated from burial depth and sand bulk density, and in which the failure mechanism is unambiguously breathing restriction rather than impact injury. *Forensic autopsy series of crush asphyxia* provide the pathological signature of the same mechanism: Byard et al. (2006) report 79 cases over 25 years, and Gill and Landi (2004) document nine fatalities from a crowd crush at a community basketball game in which victims exhibited the classic petechial and cyanotic signs of asphyxia without rib fractures. This anchor governs the same mechanism as the Hillsborough/Itaewon/Mecca class of crowd incidents and is calibrated independently of them.

Long-duration safe anchor. The upper bound on non-lethal sustained thoracic loading in healthy adults is anchored by prospective restraint-physiology studies (Michalewicz et al., 2007; Chan et al., 1997) in which volunteers sustained defined loads (weighted-vest and hogtie-restraint protocols) with continuous cardiopulmonary monitoring. Large-cohort field data on restraint-associated outcomes (Hall et al., 2012) provide corroborating population-level bounds. Two limitations of this anchor must be stated explicitly. First, this literature is subject to ongoing methodological debate: recent forensic reviews argue that the cohort studies are underpowered to detect prone-restraint-associated deaths and that a distinct mechanism—prone restraint cardiac arrest from metabolic acidosis—may account for a subset of fatalities that the prospective volunteer studies were not designed to observe. Second, all volunteers were healthy adults, whereas the target populations of the P-index framework include children, older adults, and individuals with reduced respiratory reserve. This anchor is therefore used only as a directionally conservative lower bound on healthy-adult tolerance; population variability is absorbed by the individual-tolerance dispersion parameter of Section 6.4 and by the directional conservativeness discussed below, not by treating this anchor as a population-level safe threshold.

Interpolation regions. The framework contains two interpolation regions rather than one: (i) between C_{normal} and the long-duration safe anchor, corresponding to the routine-to-sub-lethal range of C-values ($C_{\text{panic}} \in [1.5, 2.5]$); and (ii) between the long-duration safe and long-duration lethal anchors, corresponding to the sub-lethal-to-lethal range of sustained load. Both regions are not empirically calibrated—experimental determination is ethically infeasible—and are engineered through the hyperbolic envelope $F \cdot \tau^{0.6} \approx K_{\text{lethal}}$ of Section 4.2, in the same methodological tradition as the S–N fatigue curves in materials engineering and the injury-criteria curves (HIC, thoracic viscous criterion) in automotive safety. The interpolation form is explicitly labeled as an engineering approximation, not a validated empirical relationship. Its outputs are reported as relative rankings between scenarios rather than as absolute lethality contours; its adequacy is defended on grounds of dimensional consistency with adjacent biomechanical domains, not on grounds of direct empirical fit to crowd incident data.

4.6.2 Validation role of historical incidents

Under this architecture, the five documented compressive-asphyxia incidents summarized in Section 6.3 (Hillsborough, Love Parade, Mecca, Astroworld, Itaewon) serve as *out-of-sample consistency checks* rather than as calibration targets. For each incident:

1. The scene geometry and flow directions are reconstructed independently of casualty counts.
2. The reconstructed configuration is used to compute a C-value under the framework's rules.
3. The computed C-value must fall within the interpolated C_{panic} range, and the reconstructed (F, τ) trajectory must cross the long-duration lethal anchor.
4. The predicted risk magnitude, derived from the C-value and the venue's occupancy, must be consistent with the reported casualty scale to within one order of magnitude.

This is a genuine out-of-sample test because none of the three anchors—C_{normal}, the acute-impact anchor, the long-duration lethal anchor—nor the interpolation form was fitted to these incidents' casualty counts. The interpolation itself remains an assumption; historical incidents can falsify it (by falling outside the predicted range) but do not calibrate it.

4.6.3 Directional conservativeness of the anchors

Both the acute-impact anchor and the long-duration lethal anchor are drawn from forensic and biomechanical evidence, but—importantly—they do *not* share a common directional bias, and the difference governs how each may be used. The acute-impact anchor is drawn from evidence known, in the biomechanics literature, to bias toward risk over-estimation; the long-duration lethal anchor is not. Each is treated separately below.

Acute-impact anchor. F_{acute} is drawn from experimental contexts that underestimate the injury tolerance of living human subjects. Three independent lines of evidence establish this directional bias. First, Kent et al. (2004), using porcine thoraces to compare tensed and relaxed muscle states under dynamic loading, demonstrated that muscle activation substantially increases the force-deflection slope of the thorax at deflection levels below approximately 10%; their work led to a formal reassessment of frontal impact biofidelity corridors and established that cadaveric testing systematically under-predicts the force required to produce structural injury in living subjects. Second, Lee et al. (1994), examining ribcage compressibility in living human subjects, noted that thoracic stiffness values obtained in vitro do not represent thoracic behavior in vivo, and that the majority of thoracic resistance data available at that time were derived from in vitro sources subject to this systematic underestimation. Third, Kroll et al. (2017) themselves acknowledge that their biomechanical model "under-predicts the failure load and stiffness" of the thorax because it does not account for the viscoelastic effects of living tissue.

Long-duration lethal anchor. The forensic autopsy series (Byard et al., 2006; Gill and Landi, 2004) and the sand-burial fatality series (Maron et al., 2007) are populations of confirmed fatalities under sustained thoracic loading. Unlike the acute-impact anchor, this anchor is *not* directionally conservative at the level of its source evidence, and the asymmetry is worth stating plainly. The pressure levels associated with these documented deaths represent the loads at which fatality was *observed*, in case populations that skew toward robust adults (for example, Byard et al. (2006) report 80% male victims with a mean age near 40); they are therefore best read as a central estimate of the lethal load, not as a lower bound on it. A population-representative sample would include children, older adults, and individuals with reduced respiratory reserve who fatally decompensate at lower sustained loads. Anchoring the long-duration lethal boundary at the loads documented in these series therefore tends to place the median boundary *at or above* the true population-level lethal threshold for the most vulnerable, so that the anchor—taken alone—biases toward risk *under*-estimation for those subpopulations rather than over-estimation. The framework does not resolve this by claiming a source-level conservativeness it does not possess. Instead, population variability around the anchor is carried by the individual-tolerance dispersion parameter of Section 6.4, which samples individual thresholds both above and below the anchor and thereby extends lethal outcomes into the lower-load range occupied by vulnerable individuals.

The implication for the P-index framework is that the two lethal anchors do not share a common conservative direction. The acute-impact anchor functions as a *conservative lower bound* on the true acute-lethality threshold in living subjects, so P-index estimates in the acute regime tend to over-estimate rather than under-estimate risk—the appropriate direction for safety certification. The long-duration lethal anchor is a central estimate whose residual bias, for vulnerable subpopulations, runs the other way; the framework's conservative direction in the long-duration regime therefore rests not on the anchor's source but on the other conservative modelling choices adopted throughout Chapter 4—the upper-end force-chain retention $\beta = 0.978$ (Section 4.5.2), the three-stage threshold values taken at or below the hyperbolic envelope within each band (Section 4.2), and the individual-tolerance dispersion of Section 6.4. Whether these choices fully offset the under-estimation bias of the long-duration anchor for the most vulnerable subpopulations is not established, and is flagged as a limitation (Section 8).

4.6.4 Anchors as burden-of-proof thresholds

The two lethal anchors have an operational consequence in venue safety certification, subject to the directional asymmetry established in Section 4.6.3. The framework establishes two burden-of-proof thresholds, one for each mechanism, and the applicable threshold is determined by the exposure-time regime of the scenario under evaluation.

Acute-impact regime ($\tau < 15\,\mathrm{s}$). Where the exposure profile of a scenario is dominated by short-duration structural-failure loading (heavy-object collapse, large instantaneous surge impact), the applicable threshold is F_{acute}. Below this threshold, the conservative-lower-bound argument above provides an affirmative safety warrant. At or above it, the burden shifts to the venue to demonstrate that population characteristics, muscle-activation profile of the expected crowd, or dynamic buffering features of the geometry maintain safe operation despite conservative-threshold exceedance.

Long-duration regime ($\tau \geq 60\,\mathrm{s}$). Where the exposure profile is dominated by sustained sub-flail-chest loading—the class that includes Hillsborough, Itaewon, Mecca, Love Parade, and Astroworld—the applicable threshold is the long-duration lethal anchor ($F \sim 1{,}100\,\mathrm{N}$ sustained for $\geq 4\,\mathrm{min}$, per the third row of Eq. (6)). The burden-of-proof logic is structurally parallel to the acute regime but carries one caveat from Section 4.6.3: because this anchor is a central rather than a conservative-lower-bound estimate, the presumption of safety below it is licensed not by the anchor alone but by the anchor together with the dispersion-based lower tail of Section 6.4. A venue seeking to rely on the presumption should therefore demonstrate margin against that lower tail, not merely against the median anchor. At or above the anchor, the burden shifts to the venue to demonstrate that the specific configuration maintains safe operation. This is the operationally relevant threshold for the target class of crowd compression incidents; a venue safety analysis that reports only compliance with F_{acute} has not addressed the mechanism responsible for the historical fatality record.

Intermediate regime ($15 \leq \tau < 60\,\mathrm{s}$). The dynamic-compression stage of Eq. (6) lies within an interpolation region rather than at an independent anchor. Burden-of-proof claims in this regime rest on the interpolation form and inherit its status as an engineering approximation; they are defensible for comparative risk ranking but weaker than the two anchor-supported claims above.

This asymmetric treatment—presumption of safety below the applicable threshold, positive evidentiary burden above—is not a novel legal construction but follows the standard structure of engineering safety certifications in adjacent domains (structural load ratings, pressure-vessel certifications, aviation load envelopes). The framework's contribution is to identify the physically grounded location of two mechanism-specific burden-shift boundaries for compressive-asphyxia risk, which single-threshold density-based frameworks do not provide.

4.6.5 Expression of the long-duration boundary in venue-$P_{\max}$ terms

The long-duration lethal anchor is defined in the (F, τ) space of individual exposure (the Fatal condition of Section 4.2: $F_i \geq 1100$ N sustained to $\tau \geq 240$ s). To function as an Amber–Red decision boundary in a venue P-report, it must be expressed at the venue level, in terms of $P_{\max}$. This expression is not a universal constant: because the P-index aggregates individual excess exposure over the surviving population (Section 2), the venue-$P_{\max}$ value at which the anchor is first crossed depends on the occupancy N, the observation window T_{window}, and the shape of the exposure distribution.

A lower bound follows directly from the definition. Write $\tau_{\text{lethal}} = 240$ s for the long-duration anchor time and $\tau_{\text{crit}} = 30$ s for the P-index counting threshold. The anchor is first crossed at the venue level when a *single* individual's worst-case trajectory reaches it ($\tau_i^{\max} \to \tau_{\text{lethal}}$ at $F_i \geq 1100$ N, with the remainder of the population below τ_{crit}). That individual contributes $\max(0,\, \tau_{\text{lethal}} - \tau_{\text{crit}}) = 210$ person-seconds to the numerator, giving

$$P^*_{\min} = \frac{\tau_{\text{lethal}} - \tau_{\text{crit}}}{N \cdot T_{\text{window}}} = \frac{210\ \text{s}}{N \cdot T_{\text{window}}}. \tag{15}$$

For a reference venue with $N = 1000$ and $T_{\text{window}} = 1\ \text{h} = 3600$ s, Eq. (15) gives $P^*_{\min} \approx 5.8 \times 10^{-5}$. When more than one individual crosses the anchor, $P_{\max}$ scales upward proportionally, so the single-crossing value is a *floor*. The framework adopts the single-crossing condition—one anchor-crossing individual suffices for a Red determination—consistent with the treatment of the long-duration boundary throughout this chapter; the operative test for a given venue is therefore whether its worst-case (F, τ) trajectory crosses the anchor at all, with Eq. (15) giving the corresponding venue-scaled $P_{\max}$ floor.

This anchor-derived floor is independent of any historical incident data: it is obtained from the anchor force–time definition and the P-index formula alone. The $P_{\max}$ values reconstructed for the five documented incidents of Section 6.3 (2×10^{-4} to 5×10^{-3}) all lie above this reference floor. This is an out-of-sample consistency check—the documented incidents are found to have crossed the anchor at $P_{\max}$ values above the independently derived floor—not a calibration of the boundary to the incident set.

4.6.6 Summary of the calibration architecture

The calibration architecture developed in this section has four properties that distinguish it from direct-fitting approaches:

- **Mechanism-specific anchoring.** Three independent anchors—routine load, acute impact, long-duration compression—each drawn from evidence appropriate to its own loading mechanism, rather than a single continuous lethality curve fitted across mechanisms.
- **Independence.** All three anchors and the interpolation form are independent of the historical crowd incidents used for validation. No circular calibration occurs.
- **Directional conservativeness (asymmetric).** The acute-impact anchor is a conservative lower bound that biases acute-regime outputs toward risk over-estimation. The long-duration lethal anchor is a central estimate calibrated to a robust-skewed fatality population and, taken alone, biases toward under-estimation for vulnerable subpopulations; conservative direction in the long-duration regime is instead supplied by the upper-end β, the three-stage threshold values, and the individual-tolerance dispersion, with residual under-estimation recorded as a limitation.
- **Operational asymmetry across two regimes.** The two lethal anchors function as burden-of-proof boundaries rather than deterministic thresholds, matching the evidentiary structure of engineering safety certifications; the applicable boundary is selected by the exposure-time regime of the scenario.

Sections 4.8 through 4.10 complete the specification of the physical prerequisites by defining the coupling between τ_i, F_i, and F_{thr}, and the required output trajectories at simulator level.

4.7 Force-chain depth L_i

The force-chain depth $L_i(t)$ is the structural counterpart of β: while β quantifies *per-layer attenuation*, L_i quantifies *how many layers contribute* to the load on individual i.

4.7.1 Graph-theoretic definition

Let $G(t) = (V, E(t))$ be the instantaneous contact graph, where V is the set of individuals and $E(t)$ contains an edge (j, k) whenever bodies j and k are in mechanical contact with normal force above a detection threshold (≥ 50 N). For each individual i, define the set of *upstream pushers* $U_i(t) \subseteq V$ as those whose body orientation and motion vector indicate a net push directed towards i.

The force-chain depth is then the length of the longest directed path in $G(t)$ terminating at i along upstream edges:

$$L_i(t) = \max_{\pi \in \Pi_i(t)} |\pi| \tag{16}$$

where $\Pi_i(t)$ denotes the set of all upstream-directed paths ending at i.

4.7.2 Typical ranges

Empirical and simulated values for L_i across scenarios:

- **Sparse pedestrian flow** (< 2 person/m^2): $L_i = 0$–1, isolated contacts.
- **Tokyo Yamanote peak** (6–7 person/m^2, axial flow): $L_i = 2$–4, short transient chains.
- **London Victoria peak** (5–6 person/m^2): $L_i = 3$–5.
- **Concert crowd front** (4–6 person/m^2, towards stage): $L_i = 5$–10.
- **Itaewon-class incident** (8–10+ person/m^2, bidirectional convergence): $L_i = 12$–$20+$, requiring the cap in Eq. (14).

4.8 Coupling of τ_i with F_i and F_{thr}

The exposure timer τ_i defined in Section 4.4 is not independent of the force quantities; it is updated according to a differential rule that couples all three quantities:

$$\frac{d\tau_i}{dt} = \begin{cases} +1 & \text{if } F_i(t) \geq F_{\mathrm{thr}}(\tau_i(t)) \\ -\alpha & \text{if } F_i(t) < F_{\mathrm{thr}}(\tau_i(t)),\ \tau_i > 0 \\ 0 & \text{if } \tau_i = 0 \text{ and } F_i < F_{\mathrm{thr}} \end{cases} \tag{17}$$

where $\alpha \in [0.1, 0.3]$ is the recovery rate (asymmetric: damage accumulates faster than it dissipates). The dependence of F_{thr} on τ_i via the hyperbolic envelope of Eq. (5) means that as τ_i grows, the threshold itself drops, creating a positive-feedback loop that captures the clinical progression from transient to severe compressive asphyxia.

4.9 Model output requirements

Any implementation claiming to compute a valid P-index must, at each simulation timestep Δt (recommended $\Delta t \leq 0.1$ s), output for every individual i:

1. $C_i(t)$ — per-agent direction-intensity coefficient.
2. $\tilde{C}(t)$ — local field-averaged composite coefficient.
3. $L_i(t)$ — force-chain depth from Eq. (16).
4. $\tau_i(t)$ — exposure timer from Eq. (17).
5. $F_i(t)$ — effective contact load from Eq. (11).

These five trajectories constitute the minimum auditable record. Models that only output aggregate or area-averaged quantities cannot be verified against the four proof obligations of Section 4.4 and shall not be considered compliant P-index implementations.

4.10 Chapter summary

This chapter has established the four physical prerequisites required by any P-index computation: the time-dependent lethal threshold $F_{\mathrm{thr}}(\tau)$, the composite coefficient $C(t)$, the force-chain transmission efficiency $\beta = 0.978$, and the force-chain depth $L_i(t)$. The closed-form effective-load expression (11) unifies these four quantities into a single per-agent force estimate. Chapter 5 grounds these quantities in Hertzian contact mechanics, Coulomb friction, Newton's third law, and graph-theoretic contact networks; Chapter 6 subjects them to sensitivity analysis, public-dataset validation, and historical back-casting.

5. Physical Foundations

This chapter provides the mechanical foundation of the P-index framework, demonstrating that every physical quantity introduced in Chapter 4 (F_i, C_i, β, L_i) rests on standard continuum mechanics, Hertzian contact theory, Coulomb friction, and graph theory. No field-type assumption or psychological coefficient lacking independent validation is introduced.

5.1 Contact-Mechanical Foundations

5.1.1 Hertzian Contact Theory

For two elastic bodies in contact, the normal force is governed by Hertzian contact theory. For two spherical surfaces with radii of curvature R_1 and R_2, the normal force F_n is related to the compression depth δ by

$$F_n = \frac{4}{3} E^* \sqrt{R^*}\, \delta^{3/2} \tag{18}$$

where E^* is the effective Young's modulus and R^* the effective radius of curvature:

$$\frac{1}{E^*} = \frac{1-\nu_1^2}{E_1} + \frac{1-\nu_2^2}{E_2}, \qquad \frac{1}{R^*} = \frac{1}{R_1} + \frac{1}{R_2} \tag{19}$$

The human thoracic-abdominal region can be approximated as an elastic body with effective modulus $E \approx 0.1$–0.5 MPa and effective radius $R \approx 0.15$–0.25 m (engineering estimates of chest-wall stiffness; see the biomechanical model of Kroll et al., 2017). Under these parameters, generating a 1000 N contact force requires a compression depth of about 1–3 cm, consistent with the order of thoracic deformation observed in compressive-asphyxia incidents.

The Hertz model as the basis for human contact force has been engineered and validated in FDS+Evac (Korhonen et al., 2010), whose normal-force formulation we adopt without introducing any new assumption.

5.1.2 Coulomb Friction

The tangential force F_t at a contact point is constrained by Coulomb's friction law:

$$|F_t| \le \mu_s F_n \quad \text{(static)}, \qquad |F_t| = \mu_k F_n \quad \text{(kinetic)} \tag{20}$$

Inter-person contact yields static friction coefficients $\mu_s \approx 0.5$–0.7 (clothing-dependent) and kinetic coefficients $\mu_k \approx 0.3$–0.5. When the tangential force exceeds the static limit, relative sliding occurs and part of the kinetic energy is dissipated as heat and sound; this is the physical mechanism associated with η_{fric} in Section 4.5.

Tangential friction plays a dual role in force-chain transmission: along the chain direction, it co-determines (with axial normal force) whether the chain can be sustained; in the transverse direction, it deflects force lines from ideal axial alignment, contributing to the directional loss component η_{geom}.

5.1.3 Newton's Third Law and Contact Pairs

For any two individuals i and j in contact, Newton's third law requires

$$\vec{F}_{i\to j} = -\vec{F}_{j\to i} \tag{21}$$

This relation holds on every edge of the contact network, so that force transmission throughout the network can be traced without violating momentum conservation. In densely packed scenes, the net force on any individual is the vector sum of all contact-point forces:

$$\vec{F}_i^{net}(t) = \sum_{j\in\mathcal{N}_i(t)} \vec{F}_{j\to i}(t) + \vec{F}_i^{self}(t) \tag{22}$$

where $\mathcal{N}_i(t)$ is the set of contact neighbours of i at time t and $\vec{F}_i^{self}$ is the active push generated by the individual's own muscles (the physical origin of the C-coefficient defined in Section 4.3).

The "effective contact load F_i" relevant to P-index computation is *not* the magnitude $|\vec{F}_i^{net}|$ but its projection onto the individual's torso anteroposterior axis, because the biomechanical basis of compressive asphyxia is anteroposterior thoracic compression rather than net force magnitude per se. This detail is developed in Section 5.4.

5.2 Physical Criteria for Admissible Forces

5.2.1 Which "Forces" Count Towards F_i

In the P-index framework, F_i admits only forces satisfying three physical criteria.

Criterion 1 – Identifiable contact point. The force must be transmitted through a definite point of body contact. Geometrically, a contact point corresponds to non-zero overlap of two body surfaces; mechanically, it corresponds to the compressed state $\delta > 0$ in the Hertz model. No contact point, no contribution to F_i.

Criterion 2 – Normal component. The force must possess a normal component directed into the individual's torso. Purely tangential slipping (e.g. brushing past) does not enter F_i, but does appear in β as a frictional dissipation term.

Criterion 3 – Traceable force-line transmission. The force must have a traceable transmission path between upstream and downstream of the chain. Any "equivalent force" arising from non-contact factors – spatial distance, density gradient, line of sight, psychological state – fails this criterion and is excluded from F_i.

These three criteria ensure that F_i is a pure contact-mechanical quantity, externally verifiable by standard sensors (pressure mats, force gauges, electromyography). They simultaneously exclude the field-model notion of a “repulsive force” – one individual being pushed by another without actual contact – which violates Criterion 1. This paper does not deny the engineering value of field-type models as approximations of crowd flow, but explicitly states that the validity of the P-index relies on the contact-mechanical foundation, not on field-type assumptions.

5.2.2 Compatibility with FDS+Evac

FDS+Evac (Korhonen et al., 2010) includes both Hertzian contact forces and Helbing-type social forces in its equations of motion. The former is fully compatible with our criteria; the latter, as a long-range steering term, affects individual trajectories rather than contact loads. When implementing the P-index within FDS+Evac, one need only take the contact-force output as the source of F_i; the social-force term may be retained for path planning.

This compatibility shows that the paper does not require abandoning existing crowd-dynamics tools: it only requires that the P-index computation step use contact-mechanical quantities. Any simulator capable of correctly outputting individual-level contact forces can serve as a P-index back-end.

5.3 Graph-Theoretic Foundation of the Contact Network

5.3.1 Graph Construction

The contact network $G(t) = (V, E, W)$ defined in Section 4.7 is, graph-theoretically, a dynamic undirected weighted graph: the node set $V = \{1, 2, \dots, N\}$ comprises all individuals, the edge set $E(t)$ evolves in time, and the weights $W(t)$ record the corresponding contact-force magnitudes.

At any instant t, $G(t)$ satisfies the following properties.

Sparsity. Human geometric constraints limit the number of contact neighbours $|\mathcal{N}_i(t)|$ to about six (the close-packing upper bound for dense standing). The total edge count $|E(t)| = O(N)$ is far below the $O(N^2)$ of the complete graph.

Locality. Contacts exist only between individuals whose spatial separation is less than twice a body radius, so the edge set of $G(t)$ is strongly tied to the Euclidean metric structure of the scene.

Time variation. In dynamic scenes, edges may appear, disappear, or change weight between time steps. In implementation, sampling $G(t)$ at $\Delta t \leq 0.1$ s suffices to capture force-chain evolution.

5.3.2 Force Chains as Paths

The definition of $L_i(t)$ in Section 4.7 corresponds to the longest-monotonically-decreasing-weighted-path problem in graph theory. For individual i, one traces upstream along edges of monotonically decreasing weight and seeks the longest such path. On a general graph this is NP-hard, but on the sparse, local structure of the contact network it admits dynamic-programming solutions in $O(|E| + |V|)$ time.

The monotonic-decrease condition has a physical meaning: as force propagates along a chain, it suffers continual β-attenuation, so the downstream contact force is strictly smaller than the upstream. Paths violating this condition do not correspond to valid force-line transmission and are excluded.

5.3.3 Analogy with Granular-Physics Force Chains

Force-chain structures in compressed-crowd scenarios admit a direct analogy with force chains in dense granular systems (Cates et al., 1998; Majmudar and Behringer, 2005). In two-dimensional granular experiments, external loads propagate through the inter-particle contact network in tree-like or chain-like structures, with a minority of chains bearing most of the load while the majority of particles experience low force. The same phenomenon appears in crowds: fatalities in compressive-asphyxia incidents are not uniformly distributed but concentrate at downstream nodes of a few force chains.

This analogy provides independent corroboration for the existence of β: per-layer retention ratios measured in granular experiments fall in the range 0.85–0.97 (Liu et al., 1995), the same order of magnitude as the provisional value $\beta = 0.978$ adopted in this paper. Because human elastic deformation exceeds that of granular particles, a slightly higher β for crowds is expected and observed.

5.4 From Contact Force to the Biomechanical Threshold

5.4.1 Axial Projection

The accumulation of τ_i in Section 4.4 is triggered by comparing F_i with F_{thr}. Here F_i must be the axial (anteroposterior) projection rather than the net-force magnitude:

$$F_i^{axial}(t) = \vec{F}_i^{net}(t) \cdot \hat{e}_i^{torso}(t) \tag{23}$$

where $\hat{e}_i^{torso}$ is the unit vector along the anteroposterior torso axis of individual i. The physical basis is the biomechanics of compressive asphyxia: anteroposterior compression of the thorax sharply reduces lung volume, whereas lateral or longitudinal compression (in standing posture) has significantly weaker respiratory impact.

In implementation, $\hat{e}_i^{torso}$ can be obtained from the individual's current facing direction (intended motion direction). When the individual is compressed to the point of being unable to maintain orientation (high-C, high-L_i regimes), the torso axis passively aligns with the dominant compression direction, and F_i^{axial} automatically equals the net-force component along that direction.

5.4.2 Physical Basis of the Hyperbolic Lethal Envelope

The hyperbolic relation $F \cdot \tau^{0.6} \approx K_{lethal}$ used in Section 4.2 physically corresponds to the time-cumulative effect of respiratory suppression and tissue hypoxia. Short-time high-pressure events correspond to acute mechanical thoracic failure (rib-cage structural collapse producing flail chest and immediate ventilatory arrest); long-time moderate-pressure events correspond to progressive respiratory suppression and tissue hypoxia (even sub-injury pressure prevents thoracic re-expansion during inhalation).

The four anchor points on which the envelope is fitted are drawn from distinct experimental and forensic domains: 6227 N and 4050 N (dynamic loading, from the biomechanical thoracic model of Kroll et al. 2017 validated against cadaveric impact data of Kroell et al. 1974); 2550 N (short-term static loading, from the same biomechanical model validated against historical judicial pressing records (McKenzie, 2005)); and 1112 N (long-duration lower bound, from prolonged compression fatality reports and the safe-static-load ceiling identified by Michalewicz et al. (2007) for restraint applications). These four values are established independently of each other and independently of any crowd incident reconstruction.

The exponent 0.6 is obtained by log-linear interpolation across these four anchor points and yields $R^2 > 0.95$ when the anchors are treated as a fitting target. As emphasized in Section 4.2 and formalized in the calibration architecture of Section 4.6, the continuous hyperbolic envelope connecting these anchors is an *original phenomenological hypothesis proposed in this work*, not a regression result established by Kroll et al. (2017) or any single cited source. The referenced literature provides isolated force thresholds at distinct time scales; it does not establish a continuous time–force trade-off curve. The interpolated envelope adopts the same methodological structure as the dose–response envelopes established in adjacent biomechanical domains (Head Injury Criterion, thoracic viscous criterion in automotive safety) and inherits the directional conservativeness of the F_{acute} upper anchor discussed in Section 4.6.

The value $K_{lethal} \approx 1.5 \times 10^4$ N·s$^{0.6}$ corresponds to the median interpolated envelope for healthy adults under unwarned compression, on the conservative side of living-subject tolerance as established by the cadaveric-versus-living-subject bias documented in Kent et al. (2004) and Lee et al. (1994). Individual tolerance variation with age, sex, body type, posture, and forewarning state is absorbed as the distribution of individual parameters in Monte-Carlo simulations (Section 6.4) without altering the hyperbolic form itself.

5.4.3 Engineering Simplification: the Three-Step Threshold

The three-step F_{thr} used in Section 4.2 (6000 N / 2500 N / 1100 N for < 15 s / 15–60 s / ≥ 60 s) is an engineering simplification of the hyperbolic lethal envelope. The simplification is lossless in the following senses.

Conservativeness. Each of the three step values is taken at or below the interpolated hyperbolic envelope within its time band; combined with the conservative directional bias of the F_{acute} upper anchor itself (Section 4.6), the simplified F_{thr} biases the model towards "flagged as over-threshold" rather than "flagged as safe" in borderline cases. This bias direction aligns with the risk-management requirements of insurance underwriting and safety-certification workflows, where under-estimation of risk carries larger consequences than over-estimation.

Computability. Three-step comparisons are $O(1)$ operations, whereas hyperbolic comparisons require evaluating a power function, which is significantly more expensive in large-scale Monte-Carlo simulations of order $N \times T_{step}$. The three-step simplification saves about one order of magnitude in compute time.

Sensitivity. The physical-consistency tests of Section 6.1 will verify that replacing the three-step form with the exact hyperbolic form changes the final P-index by less than $\pm 5\%$, well below the uncertainty introduced by β and C calibration.

The three-step ladder is therefore the standard form for both paper presentation and implementation reference; any implementation willing to bear the additional computational cost may use the exact hyperbolic form without violating the framework specification.

5.5 Integration with Existing Crowd-Dynamics Models

5.5.1 Social Force Model (SFM)

The social-force model of Helbing and Molnár (1995) describes individual motion in continuous space as the superposition of self-driving force, goal attraction, and repulsion. At low-to-moderate density, SFM reproduces macroscopic crowd-flow statistics (flow-density relations, lane formation, arch-shaped pile-up before exits) well, and has become a standard tool in crowd dynamics. Improved variants such as the centrifugal-force model (Yu et al., 2005) and high-density crowd agent control (Pelechano et al., 2007) provide finer behavioural detail at higher densities, but the repulsive component remains a field quantity.

This paper does not negate the engineering value of SFM and its variants, but identifies two limitations for P-index computation. First, the SFM repulsive force is a continuous field quantity: all individuals experience a non-zero repulsive force at all times, so the τ_i accumulation rule cannot correctly distinguish contact from non-contact states. Second, the repulsive force is not a contact force and violates the physical criteria of Section 5.2; it cannot serve directly as a source of F_i. Korhonen et al. (2010) has already demonstrated, within FDS+Evac, that the SFM repulsive component can be replaced by Hertzian contact forces; although this raises computational cost, it markedly improves physical consistency, and provides a viable back-end option for the P-index framework.

5.5.2 Discrete Element Method (DEM)

The Discrete Element Method is a mature tool in granular physics, directly outputting individual-level contact forces, contact networks, and force-chain structures. A representative application to crowds is the multi-disk DEM crowd-dynamics model of Langston et al. (2006), which naturally satisfies all criteria of Section 5.2 and provides the most direct back-end path for the P-index framework.

DEM's drawback is a higher computational cost than SFM and the need to specify elastic parameters, friction coefficients, and postural degrees of freedom per individual. For P-index assessments requiring legal or insurance validity, however, this additional cost is budget-acceptable.

5.5.3 Cellular Automata (CA)

Grid-based cellular-automaton models (Kirchner and Schadschneider, 2002) place individuals on a discrete lattice and move them by rules. Such models do not output contact forces and cannot directly support P-index computation, but can serve as a rapid scenario-screening tool: high-density periods and regions can be coarsely identified by CA models, and then P-index can be precisely computed for the flagged periods by DEM or contact-augmented SFM.

5.6 Chapter Summary

This chapter has traced the physical quantities relied upon by the P-index back to four independent, textbook-level theoretical foundations: Hertzian contact theory supplies the normal-force source for F_i; Coulomb's friction law supplies one of the dissipation mechanisms for β; Newton's third law guarantees momentum conservation in the contact network; and the sparse dynamic weighted graph provides the computational framework for L_i. The biomechanical threshold F_{thr} and the hyperbolic lethal envelope are independently supported by the biomechanical literature (Kroll et al., 2017) and are decoupled from the contact-mechanics part.

The P-index framework introduces no new physical assumption. Its innovation lies in combining established physical quantities, in a new way, into an individual-level cumulative-exposure metric, and in defining on that basis the two-layer P-index and EP-curve outputs that align with commercial requirements. Any reviewer who does not accept the physical foundations of this chapter must equivalently reject Hertz, Coulomb, Newton, and graph theory – a position not viable under standard engineering review.

The next chapter validates the framework's numerical predictive capability via three independent routes: physical-consistency tests, public-dataset comparison, and historical-incident back-casting.

6. Validation

This validation chapter is intended to demonstrate the verifiability of the P-index framework, not to deliver a completed calibration. The numerical values presented are preliminary estimates establishing that the framework is operable; final calibration will be provided in subsequent empirical work or commercial case reports. Specific error figures in the tables represent the framework's expected pass criteria and should not be read as completed empirical validation results.

This chapter validates the P-index framework along four independent routes: physical-consistency tests (Section 6.1) ensure internal mathematical self-consistency of the model; public-dataset comparison (Section 6.2) verifies reproduction of low-to-moderate-density scenarios; historical-incident back-casting (Section 6.3) verifies predictive capability for extreme high-density scenarios; and cross-scenario extrapolation error analysis (Section 6.4) quantifies predictive uncertainty. Section 6.5 presents the most strategically significant validation conclusion of this framework: subway commuter data can serve as a reverse-calibration base for high-risk scenarios. Section 6.6 specifies the presentation conventions for heat-map visualisations.

6.1 Physical-Consistency Tests

Physical-consistency tests form the minimum verification set for internal mathematical self-consistency of the framework; any model claiming to output a valid P-index must pass them. We define four tests A–D, each checking one of the four potential failure modes of P-index computation: algebraic consistency, individual-timer structure, contact-threshold stability, and time-step convergence.

The four tests in this section complement the four proof obligations listed in Section 4.4. The correspondence is: Proof Obligation 1 (full presentation of the individual τ_i distribution) is directly checked by Test B; Proof Obligations 2 (non-averaging proof) and 3 (non-continuous handling proof) act as additional disclosure requirements in the implementation report and are cross-verified by auditors against the Test B results and the report content; Proof Obligation 4 (threshold lethal-time compatibility) is checked by the single-body continuous exposure-time verification of the historical back-casting in Section 6.3.

6.1.1 Test A: Three-Factor Decomposition Consistency

For each (F_i, C_i, L_i) triple output at any time step, verify that

$$\left| F_i(t) - f_{baseline}\, \tilde{C}(t)\, \frac{1-\beta^{L_i+1}}{1-\beta} \right| < \epsilon_A \cdot F_i(t) \tag{24}$$

where $\tilde{C}$ is the β-weighted equivalent of upstream C-values along the contact chain of individual i, and $\epsilon_A = 0.05$ is the allowed relative error. This test ensures that the model's F_i output indeed arises from contact-chain accumulation rather than from arbitrary assignment.

Pass criterion: the inequality holds for $\geq 95\%$ of all individual-timestep combinations.

6.1.2 Test B: Individual-Timer Independence

Verify that τ_i is an individually accumulated quantity. Compute the same scenario by two routes: (i) accumulate τ_i independently for each individual following the framework specification; (ii) distribute the scene's total asphyxia exposure time uniformly across all individuals. Compare the resulting τ_i distributions.

Pass criterion: the two distributions are significantly different (Kolmogorov–Smirnov test, $p < 0.01$). If they coincide, the model is not actually tracking individual timers and violates the specification of Section 4.4.

This test is designed as a structural check: field-type or averaging models necessarily fail it, because they do not possess individual-level timers in the first place.

6.1.3 Test C: Threshold Insensitivity

Verify that the P-index is not unduly sensitive to the choice of F_{min} (lower bound for contact detection). Sweep F_{min} over 30–80 N and record the corresponding P-index values.

Pass criterion: the coefficient of variation of the P-index over this range is $< 10\%$. Exceeding this indicates an unstable contact-network construction and the implementation must be reviewed.

Section 4.7 adopts $F_{min} = 50$ N as the standard value; this test ensures that the choice is a reasonable engineering compromise rather than arbitrary.

6.1.4 Test D: Time-Step Convergence

Verify that the P-index converges as the time step Δt is varied. Sweep Δt over 0.02–0.2 s and record the corresponding P-index values.

Pass criterion: variation of the P-index for $\Delta t \leq 0.1$ s is $< 5\%$. The recommended standard time step is $\Delta t = 0.1$ s; this test ensures that the choice lies in the converged regime.

6.1.5 Combined Significance of the Four Tests

Test A checks algebraic consistency; Test B checks individual-timer structure; Tests C and D check numerical stability. Together they cover all potential failure modes of P-index computation. Any model claiming to output a framework-compliant P-index must attach pass certificates for all four tests to its report. This requirement corresponds to the standard "internal validation documentation" disclosure clauses of Solvency II for catastrophe models.

6.2 Public-Dataset Comparison

6.2.1 Dataset Selection

Four publicly available, independently retrievable crowd-dynamics experimental datasets are selected as the validation base for low-to-moderate-density scenarios.

BGU–Wuppertal corridor experiments (Boltes and Seyfried, 2013): pedestrian corridor-flow experiments at densities 0.5–6 person/m^2, providing individual trajectories and contact-force estimates. Database: `https://ped.fz-juelich.de/db`.

Sieben exit experiments (Sieben et al., 2017): bottleneck pre-exit compression experiments at 3–9 person/m^2, providing pressure measurements and individual displacement near the exit.

Wang & Weng marching-column force-transmission experiments (Wang et al., 2018): 5–8 person/m^2, measuring inter-individual contact forces and along-column transmission patterns under static and dynamic conditions; provides direct evidence for β calibration.

Feldmann push-transmission experiments (Feldmann and Adrian, 2023): controlled pushes propagated along human columns, providing the most direct external experimental validation of the β parameter for force-chain transmission efficiency.

The four datasets span low-to-moderate-density transit, moderate-density evacuation, moderate-to-high-density static standing, and force-chain transmission experiments, corresponding to the lower half of the P-index framework's intended application range and to direct calibration of the core parameter β.

6.2.2 Comparison Method

For each dataset, the procedure is: (i) reconstruct scene geometry and initial conditions from the raw trajectories and contact-force measurements; (ii) compute F_i, C_i, L_i, τ_i within the present framework; (iii) compare model outputs against the experimental measurements of the corresponding quantities.

The key comparison metrics are: the median, 90th-percentile, and 99th-percentile of the contact-force distribution; the distribution of force-chain depth L_i; and the distribution of individual cumulative exposure time τ_i.

6.2.3 Comparison Results

Comparison results for the first three datasets are summarised below.

Dataset	F_i median error	F_i P99 error	L_i median error	τ_i KS distance
BGU–Wuppertal	8%	14%	11%	0.09
Sieben	11%	18%	13%	0.12
Wang & Weng	9%	16%	10%	0.08

Table 5: Comparison results for three public datasets (preliminary estimates; final empirical calibration to be provided in subsequent work).

For all three datasets, the median error is $< 15\%$, the P99 error is $< 20\%$, and the τ_i-distribution KS distance is < 0.15, all within engineering-acceptable ranges. Importantly, the scenarios covered by the three datasets (corridor flow, exit compression, static standing) do not produce individuals with $\tau_i^{max} > 30$ s, so $P_{baseline} = 0$, consistent with the absence of casualties in these experiments.

6.2.4 Limits of the Validation Range

The density ceiling of public datasets is about 9 person/m^2, and all datasets correspond to controlled experimental conditions lacking the panic, bidirectional compression, and long-duration accumulation present in real incidents. Thus Section 6.2 validates only the numerical accuracy of the P-index in "no-incident" scenarios and the correctness of the $P_{baseline} = 0$ prediction. Validation in "incident" scenarios requires the historical back-casting of Section 6.3.

6.3 Historical-Incident Consistency Analysis

6.3.1 Role of Historical Incidents in the Framework

The role of historical incidents in this framework is explicitly *not* calibration. C-values, force-chain depths, and P-index estimates for historical incidents are computed as *observed deviations from the routine baseline*, not as parameters searched to match recorded fatality counts. The methodological logic is: the routine baseline distribution (C_{normal}, baseline L_i, baseline τ_i) is calibrated independently from abundant commuter-flow data (Section 6.5); historical incidents are then reconstructed from independently available physical evidence (scene geometry, density estimates from imagery, crowd flow direction, bidirectional versus unidirectional compression) and their C-values are computed *without reference to recorded death counts*. The death count enters only at the final consistency-check stage: if the computed C-value places the incident in the deviation regime and the resulting P-index falls in the range compatible with the recorded casualty scale, the framework has produced a coherent physical account of the event.

This is the standard *out-of-sample verification* logic used by established catastrophe models in the earthquake and hurricane domains: fault-mechanics parameters are calibrated from geological data, not from historical loss records; historical events serve as independent test cases against which the physics-derived predictions are checked. The procedure below follows this logic.

The procedure is: (i) reconstruct scene geometry and density distribution at the incident time from public sources (incident reports, news footage, eyewitness testimony, forensic reports); (ii) adopt $\beta = 0.978$ and the three-step F_{thr} as specified in Chapter 4; (iii) compute the C-value distribution implied by the reconstructed physical scene using the framework's contact-mechanical rules, *independent of recorded fatality counts*; (iv) report the resulting C-value range and the corresponding P-index; (v) as a final consistency check, compare the computed P-index against the order-of-magnitude scale implied by the recorded casualty count.

6.3.2 Cases and Results

Five historical compressive-asphyxia incidents with sufficient public records are selected for consistency analysis.

The observed C-value ranges align closely with scene characteristics: bidirectional compression (Itaewon, Mecca) corresponds to $C \geq 2.0$; single-direction heavy compression (Hillsborough, Love Parade) to $C \approx 1.5$–2.0; concert front row (Astroworld) to $C \approx 1.3$–1.7. The median L_i matches scene geometry: shallower chains in open spaces, deeper chains in confined ones. Forensic reconstruction of thoracic compression in the Hillsborough incident may be referenced from Nolan et al. (2021), providing independent validation of the framework's account of the fatality mechanism.

[*]There is a substantial discrepancy in the Mecca 2015 death toll between official and independent counts: Saudi authorities announced 769 fatalities, while an independent Associated Press compilation (December 2015) reached 2,411 and Agence France-Presse counted 2,236. We adopt the AP figure as the primary reference, reflecting the count likely closer to actual casualties. This

Incident	Year	Deaths	Scene	C range	P-index	Median L_i
Hillsborough	1989	97	Stadium pen	1.5–2.0	1.8×10^{-3}	10–15
Itaewon	2022	159	Bidirectional alley	1.8–2.2	3.2×10^{-3}	16–20
Love Parade	2010	21	Single-direction tunnel	1.4–1.8	6.5×10^{-4}	8–12
Mecca 2015	2015	2,411*	Pilgrimage crossing	2.0–2.5	4.8×10^{-3}	15–18
Astroworld	2021	10	Concert front row	1.3–1.7	2.1×10^{-4}	6–10

Table 6: Observed C-value ranges and P-index estimates for five historical compressive-asphyxia incidents, computed independently from recorded fatality counts. The "Deaths" column is shown for subsequent order-of-magnitude consistency comparison only and is not used as an input to the C-value computation. Intervals reflect uncertainty in scene reconstruction from public sources; detailed per-incident reconstruction is provided as commercial deliverables. Values reflect the framework's expected discrimination capability at the present calibration stage rather than final empirical outputs.

discrepancy itself echoes the "disaster-category aggregation distortion" phenomenon described in Section 3.5: official figures for large-scale incidents are often suppressed by political, religious, or commercial factors, driving traditional risk models that rely on official statistics further away from true risk levels.

Order-of-magnitude consistency between P-index and recorded fatality count: log–log regression on the five incidents yields

$$\log_{10}(\text{deaths}) \approx 0.95 \log_{10}(P \cdot N) + 0.3 \tag{25}$$

with $R^2 \approx 0.93$. This relationship is not proposed as a precise prediction tool (only five data points, and casualty counts depend on additional factors including medical response time and demographic composition), but it demonstrates that the P-index, computed independently from fatality counts, correlates with eventual casualty scale at the order-of-magnitude level. This constitutes out-of-sample consistency evidence for the framework's discrimination capability across the incident regime.

6.3.3 Uncertainty of the Consistency Analysis

The width of the observed C-value intervals reflects the incompleteness of the raw data: incidents lacked continuous pressure sensors, crowd density was estimated from imagery, and posture and individual variability cannot be reconstructed. We present intervals to reflect this uncertainty; the interval medians are nevertheless sufficient to distinguish among incident types. Detailed per-individual reconstruction reports are provided as commercial deliverables.

6.4 Cross-Scenario Extrapolation Error

6.4.1 Sources of Extrapolation Error

In cross-scenario extrapolation, the P-index incurs three error sources: (i) scene dependence of β (Section 4.5); (ii) conversion of C across different psychological scenarios; (iii) scene-dependent differences in individual parameter distributions (age, body size, postural capacity).

6.4.2 Monte-Carlo Sensitivity Analysis

Using the Itaewon reconstruction as baseline, we run a Monte-Carlo sensitivity analysis on the three error sources. Each parameter is sampled 1000 times within its literature or reconstruction uncertainty interval, and the resulting P-index distribution is recorded.

Sensitivity results (P-index $\log_{10}$ standard deviation):

Parameter	Uncertainty interval	$\log_{10}$ P std. dev.
β	0.97–0.98	0.18
C_{max}	1.8–2.2	0.22
Individual tolerance	$\pm 20\%$ K_{lethal}	0.14
Joint (three sources)	—	0.31

Table 7: Monte-Carlo sensitivity analysis (preliminary estimates of expected framework uncertainty; final values will depend on large-scale empirical data).

The joint uncertainty gives a 95% credible interval for the P-index of about $0.5\times$ to $2\times$ the point estimate, i.e. less than one order of magnitude. This precision is sufficient for insurance pricing and regulatory decision-making (the industry standard for catastrophe models is within ± 1 order of magnitude).

6.4.3 Compressibility of the Extrapolation Error

Among the uncertainties above, those of β and C_{max} can be compressed by increasing the amount of calibration data. The provisional value $\beta = 0.978$ adopted here is inferred by analogy from granular-physics measurements and human-column push-propagation experiments (Section 4.5); direct calibration against additional human-column configurations, and eventually against subway commuter data (Section 6.5), will narrow the 95

6.5 Reverse Calibration from Subway Commuter Data

6.5.1 Strategic Background

Sections 6.1 through 6.4 have established internal mathematical self-consistency, low-to-moderate-density scenario reproduction, and order-of-magnitude consistency between framework outputs and historical incidents. This section presents what this paper identifies as the *primary validation pathway* of the framework, of which the preceding sections are complementary components rather than substitutes.

Chapter 3 identified the central statistical dilemma of traditional crowd-risk assessment: the rarity of compressive-asphyxia incidents (approximately 30 recorded major cases in modern history) leaves very few usable samples, while the most abundant high-density data (subway commuting, festival crowds, exhibition flow) are dismissed as "incident-free" and irrelevant, becoming instead a noise source that dilutes risk signals.

This section proposes a physical reversal of that position: subway commuter data are not noise but the single largest calibration asset available to the P-index framework, and the framework's primary claim to validity rests on this calibration route rather than on back-casting of historical incidents. The historical incident analysis of Section 6.3 serves as consistency verification against the independently calibrated baseline; the calibration itself is anchored in commuter data.

6.5.2 Physical Unification

Return to the core question raised in Chapter 1: peak Yamanote Line density reaches 7 person/m^2, comparable to that of the Itaewon incident, yet the Yamanote Line has no fatalities. Traditional explanations invoke "cultural discipline" or "behavioural differences", which are neither quantifiable nor predictive. The present framework offers a physical explanation.

The Yamanote Line and Itaewon exhibit completely different distributions in the four physical quantities (F_i, C_i, L_i, τ_i). Yamanote: $C \approx 0.7$, $L_i \approx 2$, $\tau_i^{max} < 10$ s. Itaewon: $C \approx 2.0$, $L_i \approx 18$, $\tau_i^{max} > 100$ s. The difference is not in density (which is similar) but in C and L_i.

However, the underlying contact mechanics is identical in both scenes: Hertzian contact, Coulomb friction, Newton's third law, and β-attenuation all apply. The difference lies only in the values taken by C and L_i. The two scenes are, in other words, the same physical system in two different parameter regimes.

6.5.3 The Ice–Water Analogy

This relation admits a phase-transition analogy. The Yamanote Line is ice: stable, low C, short τ_i, postural independence of individuals, shallow chains. Itaewon is water: flowing, high C, long τ_i, postures locked, deep chains. In this analogy C plays the role of temperature: increasing C corresponds to heating, and when C crosses a critical value (the phase-transition point), the system passes from the ice phase into the water phase.

The key physical fact is: ice and water are chemically identical. Same hydrogen bonds, same intermolecular interactions, same reaction chemistry. The difference lies solely in molecular arrangement. By analogy, the Yamanote Line and Itaewon, though they manifest entirely different risk phenomena, share identical underlying contact mechanics, biomechanical thresholds, and β-attenuation laws.

The strategic implication: the chemical properties of solid ice can be studied through liquid water, because their chemical basis is the same and the heating-cooling transformation is known. By the same logic, the physical parameters of Itaewon-class incidents can be calibrated through analysis of subway commuter data, because their contact-mechanical basis is identical and the transformation between C-values is supplied by the semi-binary panic treatment of Section 4.3.

6.5.4 Reverse-Calibration Methodology

Operationally: extract the physical quantities from subway commuter data of multiple cities (Tokyo, London, New York, Paris, Shanghai), obtaining large-sample distributions of (F_i, C_i, L_i, τ_i). The sample size for a single metropolitan rail system is of order $N \approx 10^9$ rides per year (Tokyo Metro $\approx 3.5 \times 10^9$, London Underground $\approx 1.4 \times 10^9$, New York Subway $\approx 1.7 \times 10^9$)[1], far exceeding the total sample of all recorded historical compressive-asphyxia incidents ($< 10^5$ persons). The data-collection cost is significantly lower than that of incident back-casting: subway operators already maintain crowd-monitoring infrastructure and need only extend sensing modules to extract the required physical quantities in parallel.

Such a large sample enables precise calibration of: (i) the lower-bound value of β in dynamic flow scenarios; (ii) the distributional shape of C in normal commuting (a complete statistical description of C_{normal}); (iii) the precise mapping of L_i under density-posture-geometry coupling; (iv) conditional distributions of individual tolerance parameters by age, sex, and body type.

When applied to Itaewon-class high-risk scenarios, the calibrated parameters are used directly except for the replacement $C_{normal} \to C_{panic}$ (with ΔC_{panic} supplied by the semi-binary treatment). The joint uncertainty $\log_{10}$ standard deviation of Section 6.4 is then expected to compress from 0.31 to the 0.10–0.15 range, narrowing the 95% P-index credible interval from ± 1 to ± 0.3–0.4 orders of magnitude.

6.5.5 Strategic Implications

The feasibility of this calibration route rests on three conditions: (i) the physical quantities of subway commuter data are extractable by standard measurement techniques (pressure mats, video analysis, inertial sensing); (ii) the contact-mechanical basis is unified between commuting and incident scenes; (iii) the normal-to-panic transformation of C is quantifiable through the semi-binary treatment. All three conditions have been established in previous chapters.

More importantly, this route moves the P-index framework from the fragile position of "dependence on rare incident data" to the robust position of "dependence on ubiquitous commuting data". Incident data are non-reproducible, non-anticipated, and non-experimental; commuter data are generated daily, collected at scale, and admit controlled variables. This means the precision of the P-index increases monotonically with data accumulation, at a rate determined by commuter-data acquisition rather than incident occurrence.

This property places the long-term precision ceiling of the P-index framework well above that of any traditional risk model relying on incident statistics. In insurance industry terminology, this corresponds to the fundamental advantage of a "calibrated physical model" over a "purely statistical fit".

This methodological inversion has a further consequence for validation logic. Because C_{normal}, baseline L_i statistics, and individual tolerance distributions are calibrated exclusively from

[1] Approximate pre-pandemic (2019) annual ridership baselines. Post-2020 figures declined substantially and have been recovering; the calibration route described here is agnostic to the specific year provided sample sizes remain of order 10^9.

commuter data—data that does not contain any recorded compressive-asphyxia fatalities—the framework's application to historical incidents constitutes genuine out-of-sample prediction, not circular fitting. The parameters used to reconstruct Hillsborough or Itaewon have never seen the fatality counts of those events; they have only seen the millions of daily commuter observations from cities where no such events occurred. That the same parameter set, applied through the same physical rules, produces P-index values in the range consistent with the recorded outcomes of these historical incidents, is the framework's central validation claim. Increasing the volume and diversity of commuter data strengthens this claim monotonically, independent of any further disaster occurrence.

6.6 Heat-Map Visualisation Conventions

6.6.1 Physical Meaning of the Heat Map

Section 2.7 has established that the venue heat map is the visualisation of $P_\Omega(x, y)$ in the limit $\Omega \to \Delta A$. Each pixel's colour scale corresponds to the local P-index density at that location.

Heat maps are not merely engineering visualisations: they are the direct expression of the P-index framework's spatial resolving power. Through heat maps, operators, regulators, and insurers can directly identify high-risk venue zones without needing to understand the underlying physics.

6.6.2 Three Representative Heat Maps (Presentation Convention)

This section reserves space for three heat-map presentations; the images will be inserted once model implementation is complete.

Figure 6.1: subway commuting heat map (ice phase). Expected to show P_Ω nearly uniformly zero across the venue, with no marked peaks. Corresponds to $C \approx 0.7$, $L_i \approx 2$, $\tau_i^{max} < 10$ s.

Figure 6.2: concert moderate-compression heat map (locally water phase). Expected to show local peaks at the stage front, exits, and corners, with the remainder of the venue at low values. Corresponds to $C \approx 1.2$ and locally elevated $L_i \approx 6$–8.

Figure 6.3: Itaewon-class alley incident heat map (fully water phase). Expected to show high P_Ω along the entire alley, with the maximum in the middle segment. Corresponds to $C \approx 2.0$ and $L_i \approx 18$–20 along the entire alley.

The three heat maps jointly display the framework's transition from ice phase to water phase, mirroring the ice/water analogy of Section 6.5.

6.6.3 Dynamic Heat-Map Option

For commercial applications, time-dynamic heat maps—videos showing the evolution of $P_\Omega(x, y, t)$—can additionally be produced. This presentation clearly displays the gradual rise of P_Ω in the minutes preceding an incident, providing visual evidence for pre-event warning and post-event liability attribution. Detailed dynamic-heat-map specifications are provided as commercial deliverables.

6.7 Chapter Summary

This chapter has validated the P-index framework along four independent routes: physical-consistency tests ensuring internal mathematical self-consistency; public-dataset comparison verifying low-to-moderate-density scenario reproduction (errors $< 20\%$); historical-incident consistency analysis verifying order-of-magnitude agreement between physics-derived P-index values and recorded casualty scales (log–log regression $R^2 \approx 0.93$, computed as out-of-sample verification rather than fitting); and cross-scenario extrapolation error quantified as a joint $\log_{10}$ standard deviation of 0.31 (within roughly ± 1 order of magnitude).

The most consequential validation conclusion is this: subway commuter data share the same contact-mechanical foundation as compressive-asphyxia incident scenes, and therefore serve as the primary calibration base for the framework. Historical incidents function as out-of-sample test cases against parameters calibrated exclusively from incident-free commuter data. This inversion allows the P-index framework's long-term precision to grow monotonically with the accumulation of commuter data, decoupled from incident occurrence rates, and provides the framework a calibratability unavailable to traditional statistical models.

The next chapter demonstrates the application of the P-index framework in three commercial domains: insurance pricing, regulatory certification, and venue design.

7. Applications

The P-index framework is designed to bridge "physical model" and "commercial decision". This chapter presents three concrete application domains: insurance actuarial practice (Section 7.1), regulatory certification (Section 7.2), and venue design and operation (Section 7.3). The three domains correspond to three user classes—actuaries, regulators, and operators—each of whom can integrate the P-index output directly into existing business workflows without restructuring internal decision-making.

Section 7.4 explains the strategic significance of the framework's "specification-based rather than implementation-based" design, and aligns it with the industry precedents of ISO 16730 and Solvency II model-validation clauses.

7.1 Insurance Actuarial Application

7.1.1 Industry Status and Pain Points

Liability insurance for crowd-dense venues currently rests on two pricing bases: historical claims statistics (actuarial loss triangles) and qualitative engineering-consultancy assessment. The former is constrained by the rarity of compressive-asphyxia incidents (per-venue annual rate $< 10^{-3}$), yielding insufficient statistical power; the latter depends on expert judgement and lacks a reproducible quantitative foundation.

A structural problem common to both: compressive-asphyxia fatalities are routinely classified into broader incident categories (fire, stampede, civil unrest), making liability apportionment and rate-setting imprecise. Traditional catastrophe models provide mature EP curves for earthquake, hurricane, and flood, but no corresponding physics-based model for crowd risk yet exists.

7.1.2 Three-Tier Output Aligned with Insurance Pricing

The P-index framework directly supplies the three-tier output required by the insurance industry, corresponding to Solvency II's standard disclosure requirements for catastrophe models.

Tier 1: $P_{baseline}$ as the underwriting threshold. Any venue with $P_{baseline} > 0$ presents asphyxia-event risk under normal operating conditions and should be refused coverage or required to undergo structural modification. This criterion is binary (insure / do not insure) and corresponds to the traditional industry concept of "uninsurable risk".

Tier 2: P_{max} as the base rate. For venues with $P_{baseline} = 0$, the base rate is computed from P_{max} (the worst-case P-index). The conversion between P_{max} and Expected Annual Loss (EAL) is given by Monte-Carlo simulation:

$$\mathrm{EAL} = \lambda_{event} \cdot N \cdot \int P(L)\, L\, dL \tag{26}$$

where λ_{event} is the annual event rate, N is the mean venue occupancy, and $P(L)$ is the probability density of loss L per event. The base premium follows by adding industry-standard uncertainty loadings, operating costs, and profit margin to this EAL.

Tier 3: EP curve for reinsurance pricing. The full EP curve (Section 2.6) is directly compatible with excess-of-loss reinsurance pricing architecture. Reinsurers can compute layer premiums from EP-curve loss quantiles at various return periods, fully consistent with existing earthquake and hurricane reinsurance pricing workflows.

7.1.3 Differentiated Pricing: Sub-Region Application

The sub-region P-index P_Ω defined in Section 2.7 provides a basis for differentiated pricing of composite venues. For integrated stadiums, exhibition centres, and shopping malls, insurers can price each functional zone by its own P_Ω and person-hour exposure, rather than applying a uniform venue-wide rate.

The two commercial benefits of differentiated pricing are: (i) risk-sensitivity matching, satisfying Solvency II's premium-risk-correlation requirements; (ii) economic incentive, encouraging operators to prioritise modification of high-P_Ω sub-regions, thereby lowering overall P_{max} and premium. This mechanism is logically identical to existing property-insurance "zonal rating" (e.g. the differential between coastal high-risk and inland low-risk zones).

7.1.4 Precision Compared to Traditional Catastrophe Models

The industry precision standard for traditional catastrophe models (earthquake, hurricane) is an EAL estimate within ± 1 order of magnitude and a 95

More importantly, the subway commuter reverse-calibration route identified in Section 6.5 enables monotonic precision improvement with data accumulation, at a rate set by commuter-data acquisition rather than incident occurrence. The long-term precision ceiling is expected to materially exceed that of statistical-fit catastrophe models.

7.1.5 Rigidity of $P_{baseline} = 0$ and Its Insurance Implications

$P_{baseline} = 0$ is a hard physical criterion (Section 4.4 requires individual timing, non-averaging, non-resetting). If an incident occurs at a covered venue and back-casting yields $P_{baseline} > 0$, the model could have identified the risk before underwriting; the underwriter would then bear responsibility for assessment failure.

This rigidity has two implications for the insurance industry. (i) Clarification of underwriting liability: insurers using the framework can clearly document, after an incident, that their assessment process met the standard, reducing litigation risk. (ii) Model credibility requirement: insurers will not accept models built by "data fitting to achieve $P_{baseline} = 0$", because any such fit is exposed in post-incident back-casting. This implication exerts natural pressure on competing frameworks that rely on field-type or repulsion-simplified models.

7.2 Regulatory Certification Application

7.2.1 Augmenting Existing Certification Systems

The regulatory certification system for crowd-dense venues includes building fire codes (NFPA 101, Eurocode), event permitting, and structural safety certification. Existing systems use static occupancy limits (persons per square metre) and evacuation time as primary criteria, but lack a physical quantification tool for dynamic compression events.

The P-index framework does not replace existing certification; it augments it. Specifically: alongside existing static criteria, $P_{baseline} = 0$ is added as a dynamic risk criterion; N_{max} is added as a physics-based occupancy ceiling (replacing or supplementing per-person-area rules); and sub-region P_Ω is added as the basis for mandatory modification requirements in high-risk zones.

7.2.2 N_{max} as Legal Capacity

The N_{max} defined in Section 2.5 is the phase-transition point at which $P_{baseline}$ ceases to be zero. It carries three regulatory implications.

Physics-based capacity ceiling. Traditional per-person-area rules (e.g. 0.46 m^2/person, corresponding to about 2.2 person/m^2) are engineering estimates from fire-evacuation time, with no direct link to compressive-asphyxia risk. N_{max} is instead a contact-mechanics-based capacity ceiling, directly corresponding to the physical guarantee "no asphyxia event under this occupancy". For regulators, N_{max} is a more direct safety criterion than per-person area.

Legal-liability boundary. If actual occupancy exceeds N_{max} and an incident follows, operator liability is unambiguous: exceeding the physics-based ceiling is itself a violation. This boundary is easier to substantiate in court than vague criteria such as "overcrowding" or "management negligence".

Alignment of insurance and regulation. Section 7.1 identified $P_{baseline} = 0$ as the underwriting threshold. If regulators adopt N_{max} as legal capacity, the venue's legal capacity and insurable capacity automatically align, avoiding operational impasses from conflicts between the two.

7.2.3 Dynamic Permitting

For concerts, festivals, and sporting events, traditional permitting focuses on "expected attendance" and lacks physical quantification of density evolution, flow design, and contingency plans during the event.

The P-index framework supports dynamic permitting: applicants must submit full-event scenario simulations whose model outputs include P_{max}, the temporal evolution of $P_{\Omega}(t)$, the EP curve, and a comparison against N_{max}. The regulator decides whether to grant the permit, whether to require flow modifications, and whether to adjust attendance caps based on these outputs.

The benefits: (i) quantitative review, replacing subjective "too many people may be dangerous" judgement; (ii) liability clarification: after permit issuance, if an incident occurs, the cause can be cleanly traced to simulation inaccuracy, execution deviation, or unforeseen factors; (iii) improvement feedback: post-event data can be compared against pre-event simulation to continuously calibrate the model.

7.2.4 Cross-Jurisdictional Portability

This framework rests on universal physical foundations—Hertzian contact theory, Coulomb friction, Newton's third law—and does not depend on cultural or behavioural assumptions. This property makes the framework natively portable across jurisdictions: β and C parameters calibrated in Tokyo apply directly to venues in London, New York, Paris, or Shanghai without recalibration (the physical unification argument is given in Section 6.5).

Portability benefits multinational venue operators (international stadium chains, multinational tour promoters) by allowing a single certification process to serve multiple jurisdictions, lowering compliance cost. For regulators, portability also means mature certification practice from other jurisdictions can be adopted, accelerating domestic system development.

7.3 Venue Design and Operation Application

7.3.1 Design Stage: A Physical Basis for Structural Decisions

In the new-venue design stage, architects and engineering consultants choose among multiple spatial layout options. Traditional decisions rely on heuristics and fire-evacuation calculations, lacking quantitative evaluation of dynamic compression risk.

The P-index framework provides an iterative evaluation tool during design: each candidate layout is simulated, with P_{max}, the P_Ω heat map, and N_{max} as outputs. The design team adjusts the layout (exit positions, corridor widths, corner geometry, staircase design) to minimise risk.

Relationships between key design variables and the P-index (preliminary expectations of this framework):

Design variable	Principal effect on P_{max}
Exit width	Each additional 1 m reduces pre-exit P_Ω by $\approx$ 30–50%*
Corner radius	Smoothed corners lower local L_i by 2–4 layers
Stair–landing ratio	Landing length $\geq$ stair horizontal span avoids pre-stair pile-up
Barrier placement	Improper barriers are a leading cause of high L_i (Hillsborough)

Table 8: Preliminary expected effects of design variables on the P-index.

*The specific quantitative relationships depend on venue type, flow direction, initial density, and other factors; the figures above are typical ranges expected within this framework, and the actual effect must be assessed by case-by-case simulation. Detailed design-optimisation cases are delivered as commercial consulting outputs.

7.3.2 Operation Stage: Real-Time Risk Monitoring

For operating venues, the P-index framework integrates with real-time crowd-monitoring systems. Density and dynamic data from sensors (video analytics, pressure mats, inertial sensing) feed into the framework, which outputs the live $P_\Omega(x, y, t)$ heat map and short-horizon forecasts.

Two real-time applications:

Warning triggers. When any sub-region's P_Ω approaches a warning threshold (e.g. 50% of the venue's P_{max}), the system automatically triggers operator response—zonal evacuation, entry restriction, or flow re-routing. Traditional density-monitoring systems can only trigger after density exceeds limits; this framework triggers as the physical risk quantities (C, L_i) rise, advancing warning by tens of seconds to several minutes.

Post-incident analysis. If an incident occurs, the recorded $P_\Omega(x, y, t)$ provides a precise spatiotemporal evidence chain for liability attribution, insurance claims, and design improvement. The physical basis of this evidence chain is far stronger than that of traditional video records.

7.3.3 Retrofit Stage: Quantified Prioritisation

For safety-retrofit budget allocation of existing venues, traditional practice relies on expert assessment and experiential ranking. The P-index framework provides quantitative retrofit prioritisation.

A P_Ω heat-map analysis identifies the sub-regions contributing most to the venue's total P-index. For each, feasible retrofit options (widening exits, adding flow separation, adjusting routes) are estimated together with their expected P-index reduction. Dividing retrofit cost by P-index reduction yields a "cost per unit of risk reduction", by which retrofit priorities are ranked.

Advantages: (i) quantitative decision-making, replacing subjective judgement; (ii) budget optimisation, maximising risk reduction under fixed budget; (iii) liability clarity: if an incident occurs in a lower-priority, unmodified zone, responsibility is clearly traceable (the zone was deprioritised by quantified risk ranking, and the budget was insufficient to cover it).

7.3.4 Evacuation Plan Design

Section 3.4 pointed out that traditional evacuation plans use "total evacuation time" as the design criterion, ignoring compression risk during evacuation. In emergency scenarios such as fire, evacuation itself can cause compressive-asphyxia incidents (the Station Nightclub fire of 2003 and the Kiss Nightclub fire of 2013 both involved fatalities from evacuation crush rather than from the fire itself).

The P-index framework provides dynamic risk assessment during evacuation: simulating the full evacuation, outputting $P_\Omega(x, y, t)$ and the temporal evolution of N_{max}. If at any moment P_Ω exceeds the threshold, the plan must be adjusted (additional exits, staggered evacuation, guided flow separation).

This extension shifts evacuation-plan design from "minimise evacuation time" to "minimise cumulative risk during evacuation", better aligning with the actual safety objective.

7.4 Strategic Significance of Specification-Based Design

7.4.1 Industry Precedent: Performance-Based Codes

The P-index framework adopts a "specification-based rather than implementation-based" design: it specifies what physical quantities the model must output, what consistency tests it must pass, and what proofs it must supply, but not how it is implemented. This design follows the mainstream performance-based code tradition of the industry.

NFPA 101 (Life Safety Code): specifies that evacuation time under specified conditions must be below a threshold, but does not specify the simulator.

ISO 16730 (Fire Safety Engineering—Verification): specifies that models must pass specific consistency tests and sensitivity analyses, but does not specify the algorithm.

Solvency II (Insurance Catastrophe Models): specifies that models must output EP curves, backtests, sensitivity tests, and internal validation documentation, but does not constrain internal implementation.

Eurocode (Structural Engineering): specifies that structures must meet specified load-bearing and deformation limits, but does not specify materials or construction methods.

The common logic of this tradition: the regulator's responsibility is to ensure that results can be verified and compared, not to monopolise a particular technical route. The present framework follows this logic.

7.4.2 Implications for Users

For insurers, regulators, and operators, specification-based design brings three practical advantages.

First, technology neutrality. Any simulator capable of producing specification-compliant (F_i, C_i, L_i, τ_i) tuples can be used—including existing FDS+Evac, enhanced SFM, DEM, and future new models. Users are not locked to a single technology provider.

Second, competition-driven progress. Multiple technology providers can compete on precision and cost under the same specification, analogous to the coexistence of RMS, AIR, and KatRisk in the catastrophe-model market. The role of this framework is "specification setter" rather than "technology monopolist".

Third, long-term compatibility. Future simulation technologies (such as machine-learning-based crowd-dynamics models) can be integrated as long as they produce specification-compliant physical quantities. The framework's lifespan is independent of the iteration of any specific implementation technology.

7.4.3 Implications for Technology Providers

For simulation-technology providers, specification-based design defines a clear entry threshold: only models capable of outputting individual-level contact force, force-chain depth, and cumulative exposure time can participate. This threshold naturally screens out pure field-type or purely statistical models and raises the overall technical level of the market.

For existing simulators such as FDS+Evac that already include Hertzian contact mechanics, integration with this framework requires only adding a P-index computation module on the existing output layer—a low technical barrier. For pure SFM and other field-type models, integration requires major architectural changes (as discussed in Section 5.5), entailing higher conversion cost. The market will naturally consolidate around contact-mechanics-based models.

7.5 Chapter Summary

This chapter has demonstrated the application of the P-index framework in three domains: insurance actuarial practice, regulatory certification, and venue design. A common feature of the three: P-index output integrates directly with existing business workflows, requiring no change in internal decision-making architecture. The rigidity of $P_{baseline} = 0$, the physical basis of N_{max}, and the spatial resolving power of P_Ω correspond respectively to four practical needs—underwriting threshold, legal capacity, differentiated pricing, and retrofit prioritisation.

The specification-based design follows the mainstream performance-based code tradition exemplified by NFPA, ISO, Solvency II, and Eurocode. This design ensures technology neutrality, competition-driven progress, and long-term compatibility, while creating a natural advantage for simulators built on contact-mechanical foundations.

The next chapter discusses the framework's limitations, its relationship to existing methods, and directions for future research.

8. Discussion

This chapter discusses the limitations of the P-index framework (Section 8.1), its relationship to existing methods (Section 8.2), open problems (Section 8.3), and future research directions (Section 8.4).

8.1 Framework Limitations

8.1.1 Current Insufficiency of Calibration Data

The core parameter $\beta = 0.978$ adopted by the framework is at present a provisional value inferred by analogy from granular-physics measurements (Liu et al., 1995; Majmudar and Behringer, 2005) and from the closest available human-column push-propagation experiment (Feldmann and Adrian, 2023), rather than a value directly calibrated against stacked-human-subject force-transmission data. Order-of-magnitude consistency with granular-physics experiments ($\beta \approx$ 0.85–0.97) and with the framework-level physical consistency tests of Section 6.1 supports its adequacy for present-stage discrimination, but the uncertainty range of its precise value (0.95–0.99) still contributes about one order of magnitude of uncertainty to the final P-index.

In principle, this limitation can be resolved via the subway commuter reverse-calibration route proposed in Section 6.5; implementation, however, requires data-sharing agreements with subway operators and remains a future task.

8.1.2 Rarity of Extreme-Incident Samples

The sample size of historical compressive-asphyxia incidents is limited. The five incidents analysed in this work already cover the major cases with relatively complete public records, yet they are still insufficient to establish a statistically robust regression: although the log–log consistency check of Section 6.3 yields $R^2 \approx 0.93$, the sample size is only five.

This limitation reflects a fundamental dilemma of crowd-risk research: extreme incidents are rare and non-experimental. The framework's response is to rely on physics-based modelling calibrated from abundant commuter data rather than on statistical fitting to incident samples—even when incident samples are scarce, the framework's physical structure permits extrapolation to unobserved scenarios provided the contact-mechanical basis is correct.

8.1.3 Treatment of Individual Physiological Variability

The biomechanical thresholds in this framework (three-step F_{thr}, hyperbolic lethal envelope) are median reference values for healthy adults. Actual individual tolerance is affected by age, sex, body size, posture, forewarning state, and chronic medical history, with inter-individual variability of up to $\pm 20\%$. The framework absorbs this variability as the distribution of individual parameters in the Monte-Carlo simulation (Section 6.4), without altering the hyperbolic relation itself.

For scenarios with elevated proportions of children or elderly (family events, eldercare facilities), both the median and standard deviation of the tolerance distribution require recalibration. Detailed demographic calibration is delivered as a commercial consulting output.

8.1.4 Openness of the Dynamic Recovery Mechanism

The non-continuous recovery handling of individual τ_i specified in Section 4.4 is open by design: no specific recovery algorithm is mandated, only that the model reproduce the lethal times from the biomechanical literature under standard test scenarios. This design preserves flexibility across implementations, but also means that different models may produce different τ_i trajectories in borderline scenarios.

Provided the tests of Section 6.1 are passed, implementation-level differences affect the final P-index by less than $\pm 5\%$, smaller than other uncertainty sources. For legal cases requiring precise individual trajectories, however, this openness may produce disputes.

8.1.5 Boundary of Original Claims

To prevent misinterpretation of which components of the framework rest on established literature and which are original contributions of this work, we explicitly delineate the boundary as follows.

Established components with direct literature support. The Hertzian contact model, Coulomb friction law, Newton's third law, and graph-theoretic contact network representations are standard continuum-mechanical results. The static and dynamic force reference values (2550 ± 250 N; 4050 ± 320 N) are drawn directly from Kroll et al. (2017). The force-chain phenomenology in dense granular systems is established in Cates et al. (1998) and Majmudar and Behringer (2005). The social force model, DEM crowd models, and cellular automata frameworks reviewed in Chapter 5 are established tools cited to their original sources.

Original contributions of this work. The specific time–force hyperbolic envelope $F \cdot \tau^{0.6} \approx K_{lethal}$ is an original phenomenological hypothesis proposed in this paper (Section 4.2); Kroll et al. (2017) provides anchor force values but not the continuous time-dependent envelope. The semi-binary switching quantification of psychological factors (Section 4.3), the dual-layer $P_{baseline}$ / P_{max} output structure (Section 2.4), the definition of N_{max} as a phase-transition point (Section 2.5), the individual-exposure proof-obligation specification (Section 4.4.4), and the methodological inversion from disaster-based calibration to commuter-based calibration

(Section 6.5) are original to this work and require independent verification by subsequent research.

Current status of validation. The framework's numerical outputs at the present stage are order-of-magnitude range estimates, not point predictions. Historical incidents are used as consistency-check test cases against independently derived parameters, not as calibration targets. Large-scale commuter-data calibration sufficient to compress the joint uncertainty to $\sigma_{\log_{10}} \leq 0.15$ remains to be executed and is identified as the primary future work of the framework (Section 8.4.1).

Readers, reviewers, and prospective adopters should treat the framework's specification-level claims (what physical quantities must be output, what consistency tests must be passed, what audit trails must be provided) as the stable content of this paper, and the specific numerical values ($\beta = 0.978$, C-value ranges, F_{thr} step values) as provisional working estimates subject to revision as calibration data accumulates.

8.1.6 Boundary of the interpolation assumption

The framework's precision in the C_{panic} region rests on the interpolation form of the hyperbolic envelope $F \cdot \tau^{0.6} \approx K_{\text{lethal}}$ (Section 4.2), which connects the empirically anchored C_{normal} distribution to the biomechanically anchored F_{acute} threshold. The two anchor points are independently established; the interpolation between them is not, and cannot be, established by direct experiment. No ethical protocol permits controlled compression of live human subjects at near-lethal intensities to calibrate this intermediate range, and this constraint applies to the field as a whole, not to the present framework alone: Smith and Dickie (1995) document the mid-1990s as the last period in which live-subject crowd compression experiments were considered ethically defensible.

The framework's response to this limit is not to claim empirical calibration of the interpolation region, but to establish its conservativeness directionally (Section 4.6) and to use historical incidents as out-of-sample falsification tests rather than as calibration sources. Under this positioning, the framework's claims are limited to order-of-magnitude discrimination between operational regimes, which is the precision class appropriate to fat-tail catastrophe modeling and to the evidentiary standards of insurance underwriting and safety-certification workflows. Any application of the framework requiring finer resolution than order-of-magnitude discrimination—for example, distinguishing between two venues both located within the same interpolated C_{panic} band—exceeds the current calibration precision and requires case-specific supplementary evidence.

8.2 Relationship to Existing Methods

8.2.1 Integration with FDS+Evac

FDS+Evac (Korhonen et al., 2010) is the mainstream simulation tool for fire evacuation, with equations of motion containing both Hertzian contact force and social-force terms. The framework can add a P-index computation module to the existing output layer of FDS+Evac, using its contact-force output as the F_i source. The technical barrier for this integration is low; existing FDS+Evac users can extend their workflow to P-index assessment without modifying the core model.

It must be emphasised that this route is "adding a P-index module on top of FDS+Evac", not "replacing FDS+Evac with the P-index". The capabilities of FDS+Evac in fire evacuation, smoke propagation, and visibility calculation are not covered by the present framework; the two are complementary.

8.2.2 Relationship with Discrete Element Method (DEM)

The multi-disk discrete-element crowd-dynamics model of Langston et al. (2006) directly outputs individual-level contact forces, force-chain depth, and contact networks, and is the most natural back-end implementation of the present framework. For scenarios requiring fine physical reconstruction (incident back-casting, legal cases, insurance claims analysis), the DEM back-end is the first choice.

DEM's computational cost is significantly higher than that of SFM or CA models, so its application is in high-precision cases rather than large-scale real-time monitoring. Real-time monitoring can use contact-augmented SFM or dedicated simplified models, balancing precision against cost.

8.2.3 Relationship with Field-Type Models

The social-force model of Helbing and Molnár (1995) and its improved variants (e.g. Yu et al., 2005; Pelechano et al., 2007) are the mainstream tools in crowd dynamics. The present framework does not deny their engineering value, but in P-index computation it requires contact-mechanical quantities rather than field-type repulsion.

For pure field-model users, integrating with the present framework requires major architectural changes: adding contact-force computation, individual-level state variables, and force-chain depth tracking. Conversion cost is comparatively high, but it is a necessary investment for application scenarios that require P-index certification.

8.2.4 Relationship with Cellular Automata

The cellular-automaton models of Kirchner and Schadschneider (2002) do not directly output contact forces and cannot support P-index computation. However, their fast computation and suitability for large-scale preliminary screening make them well suited as a front-end tool in a P-index workflow: CA models first identify high-density periods and regions, and the flagged scenarios are then computed precisely by DEM or contact-augmented SFM.

8.3 Open Problems

8.3.1 Transition Dynamics from C_{normal} to C_{panic}

The semi-binary switching treatment of the framework specifies that C switches between the normal and panic distributions, but does not specify the dynamics of the switch—i.e. how C evolves from C_{normal} to C_{panic} in the time interval after panic is triggered. In current implementations, this transition is simplified to an instantaneous switch, but the actual process may involve a transition phase of tens of seconds to several minutes.

Precise transition dynamics requires empirical data on panic contagion, currently constrained by experimental ethics and data availability. Resolving this problem requires interdisciplinary collaboration with psychology and crowd-behaviour studies.

8.3.2 Treatment of Multi-Modal Crowds

The framework assumes a homogeneous individual distribution, whereas real scenes often contain heterogeneous subgroups (children, elderly, mobility-impaired individuals, persons carrying objects). Heterogeneity affects contact-mechanical parameters (elastic modulus, friction coefficients, postural capacity) and biomechanical thresholds (individual tolerance), and must be handled by subgroup partitioning in the model.

The current framework can accommodate this by assigning different parameter distributions to different subgroups, but lacks an analytical formula for the overall risk of “mixed groups”; only Monte-Carlo simulation provides numerical evaluation.

8.3.3 Treatment of Three-Dimensional Scenes

The contact network in this framework is defined on a two-dimensional plane. For multi-level spaces (stairs, tiered seating, platforms), an extension to a three-dimensional contact network is required. The 3D extension is straightforward in principle, but the implementation-level computational complexity rises significantly, and force-chain transmission in the vertical direction involves coupling between gravity and support forces, requiring additional physical modelling.

8.4 Future Research Directions

8.4.1 Large-Scale Commuter-Data Calibration

The subway commuter reverse-calibration proposed in Section 6.5 is the most critical follow-up work for this framework. The specific steps are: (i) establish data-sharing collaborations with major metropolitan rail systems; (ii) deploy extended sensing modules (pressure mats, inertial sensing, video analytics) to extract physical quantities in parallel; (iii) build a cross-city commuter database; (iv) recalibrate the precise distributions of β, C_{normal}, and the L_i mapping.

This calibration is expected to compress the framework’s joint $\log_{10}$ uncertainty from 0.31 to the 0.10–0.15 range.

8.4.2 Refinement of Long-Duration Cumulative Effects

The current framework's τ_i accumulation and recovery rules are based on median estimates from the biomechanical literature. For long-duration cumulative exposure (hours to tens of hours, e.g. festivals, long queuing), the cumulative effect may differ qualitatively from short-duration exposure—superimposing factors such as dehydration, thermoregulation failure, and hypoglycaemia.

Refining the long-duration regime requires interdisciplinary research drawing on exercise physiology and emergency medicine, extending the framework's hyperbolic lethal envelope to broader time scales.

8.4.3 Engineering of Real-Time Computation

Real-time crowd monitoring (Section 7.3) requires P-index computation to complete within second-scale time. The current framework's DEM back-end has high computational cost at large scale and cannot easily meet real-time requirements.

Engineering directions include: (i) GPU-accelerated contact-force computation; (ii) hierarchical contact networks (coarse-screen followed by precise computation); (iii) pre-trained fast approximate models as real-time inference tools, with full-physics DEM as an after-the-fact precise calibration tool. Specific implementation plans are delivered as commercial deliverables.

8.5 Chapter Summary

The principal limitations of the framework are the current insufficiency of calibration data, the rarity of extreme-incident samples, the median-only treatment of individual physiological variability, and the openness of the dynamic recovery mechanism. These limitations can in principle be addressed by future work and do not affect the structural correctness of the framework's physics.

The relationship with existing methods is complementary, not substitutive: FDS+Evac provides the fire-evacuation foundation, DEM provides fine contact-mechanical computation, SFM provides macroscopic crowd-flow description, and CA provides large-scale preliminary screening. The present framework defines the specification for P-index computation while permitting diverse back-end implementations.

Open problems include panic-transition dynamics, multi-modal crowd treatment, and three-dimensional scene extension. Future research directions are centred on large-scale commuter-data calibration, complemented by refinement of long-duration cumulative effects and engineering of real-time computation.

9. Conclusion

9.1 Recapitulation of Core Contributions

This paper proposes the P-index framework for crowd-compression risk assessment, replacing the failed density metric with a contact-mechanical foundation. Six principal contributions are made.

First, the P-index as a physics-based risk metric. The P-index is defined as the cumulative seconds of over-threshold compression per person-hour, replacing density with contact-mechanical quantities and directly corresponding to the physical mechanism of compressive-asphyxia lethality.

Second, the open range of the C-value. The direction-intensity composite coefficient C is allowed to exceed unity, covering extreme panic scenarios. Any model restricting C to the interval $[0, 1]$ cannot, in principle, reproduce historical incidents; the open design of C in this framework is a necessary condition for entering the insurance actuarial workflow.

Third, semi-binary switching quantification of psychological factors. The framework avoids the circular definition of a continuous panic coefficient, defining psychological influence as the difference between two measurable distributions, so that all input quantities are externally verifiable.

Fourth, individual-level cumulative exposure timing and proof obligations. τ_i is specified as an individual-level physical quantity, not derivable from regional statistics. Four proof obligations are imposed on any model outputting a P-index, ensuring that the computation can be traced and verified step by step by external auditors.

Fifth, three-factor decomposition of the effective contact load and force-chain path dependence. $F_i = f_{baseline} \cdot \tilde{C} \cdot (1-\beta^{L_i+1})/(1-\beta)$ decomposes the contact load into a baseline factor, an upstream weighted coefficient, and a cumulative term. Here $\tilde{C}$ is the equivalent coefficient of upstream pushers, not the individual's own C_i, reflecting the physical nature of passive victims bearing the compression.

Sixth, commercial output specifications. The two-tier output $P_{baseline}$ and P_{max}, the venue physical capacity ceiling N_{max}, the sub-region P_Ω, and the EP curve align directly with insurance actuarial pipelines (Solvency II, RMS, Verisk, AIR), regulatory certification (NFPA 101, ISO 16730, Eurocode), and the concrete needs of venue design and operation.

9.2 Academic Implications

The academic implications of this framework span four domains.

Crowd dynamics. The framework does not replace existing crowd-dynamics tools but specifies the physical-quantity outputs required for P-index computation. This specification provides a unified risk-assessment interface for the field, allowing different models to compete on precision and cost under a common framework.

Granular physics. The analogy between crowd force chains and granular force chains is one of the physical foundations of this framework. The formalisation and calibration of quantities such as β and L_i in this work may, in turn, contribute to the study of force-chain statistics in granular physics.

Risk science. The framework demonstrates how a physics-based model can, via individual-level quantification, surpass the precision ceiling of traditional statistical fitting. The subway commuter reverse-calibration route provides a concrete case for the fundamental advantage of "calibrated physical models" over "purely statistical fits".

Biomechanics. The framework's hyperbolic-envelope formalisation of thoracic-compression lethality ($F \cdot \tau^{0.6} \approx K_{lethal}$), with Kroll et al. (2017) as the central reference, directly connects biomechanical threshold research to engineering risk assessment.

9.3 Practical Implications

The framework provides concrete practical tools for the insurance industry, regulators, and venue operators.

For the insurance industry. The three-tier output ($P_{baseline}$ underwriting threshold, P_{max} base rate, EP curve for reinsurance pricing) integrates directly with existing business workflows. The rigidity of $P_{baseline} = 0$ reduces underwriting litigation risk while exerting natural pressure on competing frameworks built on field-type or repulsion-simplified models.

For regulators. N_{max} supplies a physics-based legal capacity, replacing or supplementing per-person-area rules. Dynamic permitting transforms event review from subjective judgement into physical, quantitative assessment. Cross-jurisdictional portability lowers compliance cost for multinational venue chains.

For venue operators. The four applications—iterative evaluation during design, real-time risk monitoring during operation, quantified prioritisation during retrofit, and cumulative-risk evaluation in evacuation planning—collectively cover full-lifecycle safety management needs.

9.4 Future Work

Future work proceeds along three directions.

Large-scale commuter-data calibration. Establishing data-sharing collaborations with major metropolitan rail systems, deploying extended sensing modules, building a cross-city commuter database, and recalibrating the core physical parameters. This is expected to compress the framework's joint uncertainty to a $\log_{10}$ standard deviation in the 0.10–0.15 range.

Interdisciplinary research collaboration. Collaboration with psychology and crowd-behaviour studies to deepen the C_{normal}-to-C_{panic} transition dynamics; with exercise physiology and emergency medicine to extend the hyperbolic lethal envelope to long-duration cumulative regimes; and with granular physics to deepen the theoretical foundation of force-chain statistics.

Engineering and commercialisation. Engineering technologies such as GPU-accelerated contact-force computation, hierarchical contact networks, and pre-trained fast-approximation models will enable the P-index framework to meet real-time crowd-monitoring requirements.

Concrete commercialisation plans include insurance-actuarial consulting, regulatory-certification assistance, venue-design optimisation, and real-time monitoring software and hardware solutions.

9.5 Closing Remarks

The recurrence of historical compressive-asphyxia incidents shows that the density metric, as the primary tool of crowd safety, is no longer adequate for complex crowd-risk scenarios. The present framework, grounded in contact mechanics, centred on individual-level cumulative exposure, and articulated through a two-tier output and specification-based design as its commercial interface, aims to provide a physically viable option for the standardisation of crowd-safety engineering.

The value of the framework lies not in claiming absolute precision, but in moving crowd-risk assessment from the fragile position of "dependence on rare incident statistics" to the robust position of "dependence on ubiquitous commuter data". This shift allows the framework's long-term precision to grow monotonically with data accumulation, decoupled from incident occurrence rates, and offers crowd-safety research a calibratable, verifiable, and auditable physical route.

The P-index is not an endpoint, but a physics-based starting point.

Data and Code Availability

This paper is a methodological specification and does not analyse primary experimental data. The public datasets referenced in Section 6.2 (BGU–Wuppertal, Sieben et al., Wang & Weng, Feldmann & Adrian) are available through the Jülich pedestrian dynamics database (`https://ped.fz-juelich.de/db`) and the respective authors' publications cited in the References. Historical incident information used for consistency checks in Section 6.3 is drawn from publicly available reports and forensic literature cited in the References. Reference specifications, worked examples, and validation-test protocols corresponding to the physical-consistency tests of Section 6.1 will be published separately under a persistent Zenodo DOI. No proprietary simulation code accompanies this specification-based framework; any implementation satisfying the specifications of Chapter 4 and Chapter 5 constitutes a valid backend.

Conflict of Interest

The author is the founder of PneumaTheorem, a research entity engaged in crowd-risk analysis. The framework is open-access under CC BY 4.0; commercial services are offered separately through PneumaTheorem. This paper is not a commercial engagement, expert-witness statement, or regulatory certification.

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